MCIMX8QM-CPU MEK Platform

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ICAP Classification:	CP: ____	IUO: ____	PUBI: X
Drawing Title:	i.MX 8QM CPU CARD		
Page Title:	FRONT PAGE		
Size: A2	Document Number: SOURCE: SCH-29420, PDF: SPF-29420		Rev: H1
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Schematic capture done with Allegro Design Entry CIS 17.2. Schematic also available in 16.2.

i.MX 8QM CPU Card

Part Number: MCIMX8QM-CPU

This board was designed for maximum flexibility in software development and demonstrates multiple functions possible with i.MX processors. Although best design practices have been applied, some areas may not be suitable for a mass production design. For an added resource, refer to Hardware Development Guide document.

Consumer devices were utilized in this design when lead time for equivalent automotive-grade devices conflicted with production schedules. NXP suggests consulting component suppliers for equivalent automotive-grade device information.

DNP appearing near a component signifies "do not populate." These parts are not installed

Additional information on compatible daughter cards, cameras, etc. is provided on the nxp.com website.

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11	MX8QM GND
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17	BOOT CONFIGURATIONS & SPI FLASH
18	RESET & LEDs & JTAG
19	DEBUG UART TO USB
20	USB3.x TYPE C
21	PCIe & SATA
22	HDMI TX & RX
23	ETHERNET PHY
24	AUDIO CODEC WM8962
25	WIFI & BLUETOOTH
26	SENSORS
27	LVDS CONNECTORS
28	MIPI CSI CONNECTORS
29	MIPI DSI CONNECTORS
30	BOARD TO BOARD CONNECTORS
31	LEVEL TRANSLATORS & DRIVERS
32	MISCELLANEOUS

LVDS TO HDMI Daughter Card
Schematic SCH-29680
Part No. IMX-LVDS-HDMI

MIPI TO HDMI Daughter Card
Schematic SCH-29678
Part No. IMX-MIPI-HDMI

i.MX 8QXP CPU Card
Schematic SCH-29683
Part No. MCIMX8QXP-CPU

i.MX 8QM CPU Card
Schematic SCH-29420
Part No. MCIMX8QM-CPU

Common Base Board
Schematic SCH-29918
Part No. MCIMX8-8X-BB

ENET Daughter Cards

Audio IO Card
Schematic SCH-29941
Part No. IMX-AUD-IO

REV	Revision Notes	Date
C1	1. Sheet No. 8 : Pull-Up Resistors removed from PMIC pin 1. Pin 44 tied to Pin 41, to minimize quiscent current 2. Sheet No. 17 : SATA Boot Option removed from the boot mode list 3. Sheet No. 22 :1uF Capacitor is added in series with HDMI_RX0_ARC_P (For fixing HDMI RX side issues) 4. Sheet Nos. 9 & 16: Changed R383 from 1K to 100E, SD Card Power Changed from VCC_PER_3V3 to VCC_EXT_3V3 and Discharge Circuit for EXT_1V8 Added (For fixing power discharge issues while Processor reset) 5. Sheet No. 24 : Changed Headphone Jack Pinout to AHJ (Pins 1 & 4 Swapped) and added AHJ Graphic (Headphone JacK reconfiguration) 6. Sheet No. 19 : FTDI Chip updated to FT4232H 7. Sheet No. 30 : Connector J20 symbol updated 8. Sheet Nos.12 & 19 : Option provided for connecting SCU UART signals to FTDI Chip for debug 9. Sheet No . 21 : PCIe clock selection (internal / external) option provided 10. Sheet No 9 : PMIC2 WDI disconnected as per PMIC errata ER023 (R113 Unmounted) 11. Sheet Nos 14 & 15 : DDR_CH0_RST_B & DDR_CH1_RST_B Pulldown to Ground with 10K (Added R1481 & R1482) 12. Sheet Nos 12: Added R1483 & R1484 for ANA_TEST_OUT 0 and 1	08-08-2018
C2	1. Processor part number updated to "PIMX8QM6AVUFFAB"	17-09-2018
C3	C3 is internal release, not used for layout update. 1. PMIC_1 (U10) P/N updated to MC33PF8100EPES PMIC_2 (U23) P/N updated to MC33PF8100EQES 2. Following obsolete P/N updated:- DA1,DA2 - BAV99LT1G (ON SEMICONDUCTOR) J1,J6 - 47659-1100 (MOLEX) U17,U18 - MT53E768M32D4DT-053 AIT:E (MICRON)	22-Nov-2019
C4	No electrical changes. 1. Classification changed to Public Information. 2. Note updates. 3. U15 Processor and U10/U23 PMICs updated to production part numbers. 4. Following P/N updated back to: DA1,DA2 - BAV99 (FAIRCHILD) J1,J6 - 47659-1000 (MOLEX)	24-Jan-2020
C5	Following Parts Updated:- 1. HDMI Connector J1 and J6 updated to 476591100. Footprint number differs from layout rev C1, but is compatible per CM. 2. U37 updated to PTN5110NTHQZ. 3. DA1 and DA2 updated to BAV99LT1G. Footprint number differs from layout rev C1, but is compatible per CM. 4. C531 made as DNP 5. R369 value changed to 23.2K for increasing Current limit of load switch to 3A	09-Jun-2020
C6	No electrical changes. 1. Processor does not support MLB; removed MLB from diagrams on sheets 3 & 4. Added notations on sheets 6, 10, 13, 16, 30. 2. Added USB VBUS disable pull-down note & VBUS circuit (as a note) to sheet 20. 3. Added UART3 & UART4 net name error notes to sheets 13 & 30.	13-Jan-2021
C7	No electrical changes. 1. Fixed BOM notation error on C547. The assembly option was "NM" and now changed to a blank field. This capacitor has always been installed. See sheet 20.	18-Jan-2021
D/E	1. List of Components Changed:-	1-Nov-2022 11-Sept-2023

List Of Components Changed

Ref Des	Previous Version	New Version
U20	ISL29023	VCNL4035X01-GS08
U52	IR3827MTRPBF	LTB640EUDC-1#PBF
U53	MTFC32GAKAEEF-AIT	MTFC32GAZAQHD-AIT
D34	MBR0835LG	MBR0835L
U29	WM8960CGEFL/RV	WM8962BECSN/R
Q6	FDMS6681Z	NTMFS05P03P8ZT1G
C20	1210X107K100SNT	LMK325B107MM-T
D12	MBR1405FT1G	MBR1405FT
U13	FXAS21002CQ	I3G4250D
U14	MPL3115A2	MPL3115A2R
U49, U50	IR3895MTRPBF	LTB640 [315-80918]
U37	PTN5110NTHQZ	PTN5110NHQZ
U21	FXOS8700CQ	LSM303AGR

2. Added Test Points for the Aux +/- pins.

REV	Revision Notes	Date
F	Replaced 1.6G part with 1.3G part i.e. Replaced U15 from MIMX8QM6AVUFFAB to MIMX8QM6AVUFEAB.	25-Sept-2024
G	Added Board Edge to paste WiFi Antenna.	10-APRIL-2025
H	DNP-R8 and R9 Replaced C5 and C7 with 0 ohm (R1877and R1878)	21-AUG-2025
H1	Changed the Revision Text to H1. No Netlist or BOM change.	03-SEPT-2025

ICAP Classification: CP: ___ IUO: ___ PUBI: X

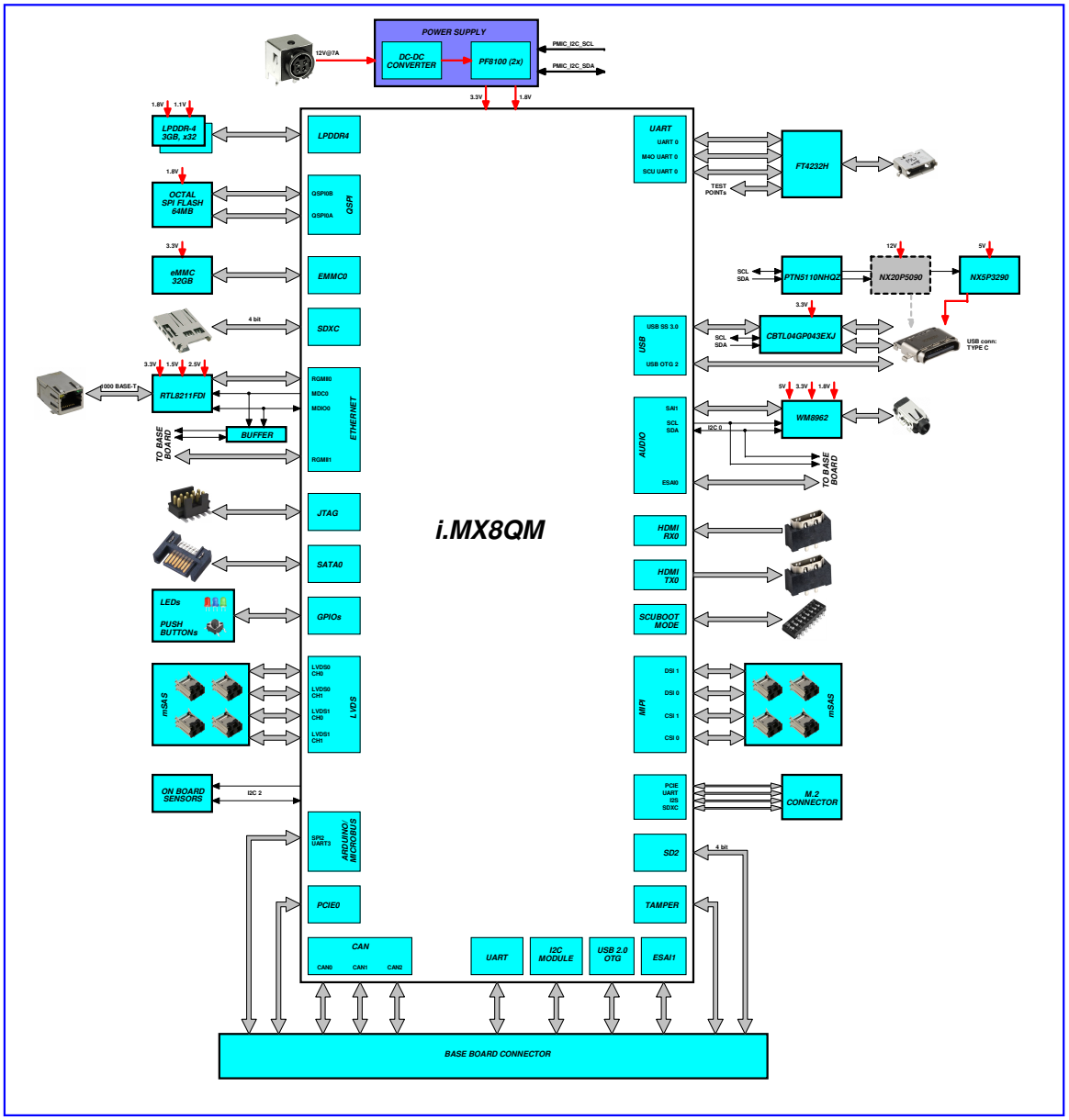
Drawing Title: i.MX 8QM CPU CARD

Page Title: TITLE PAGE

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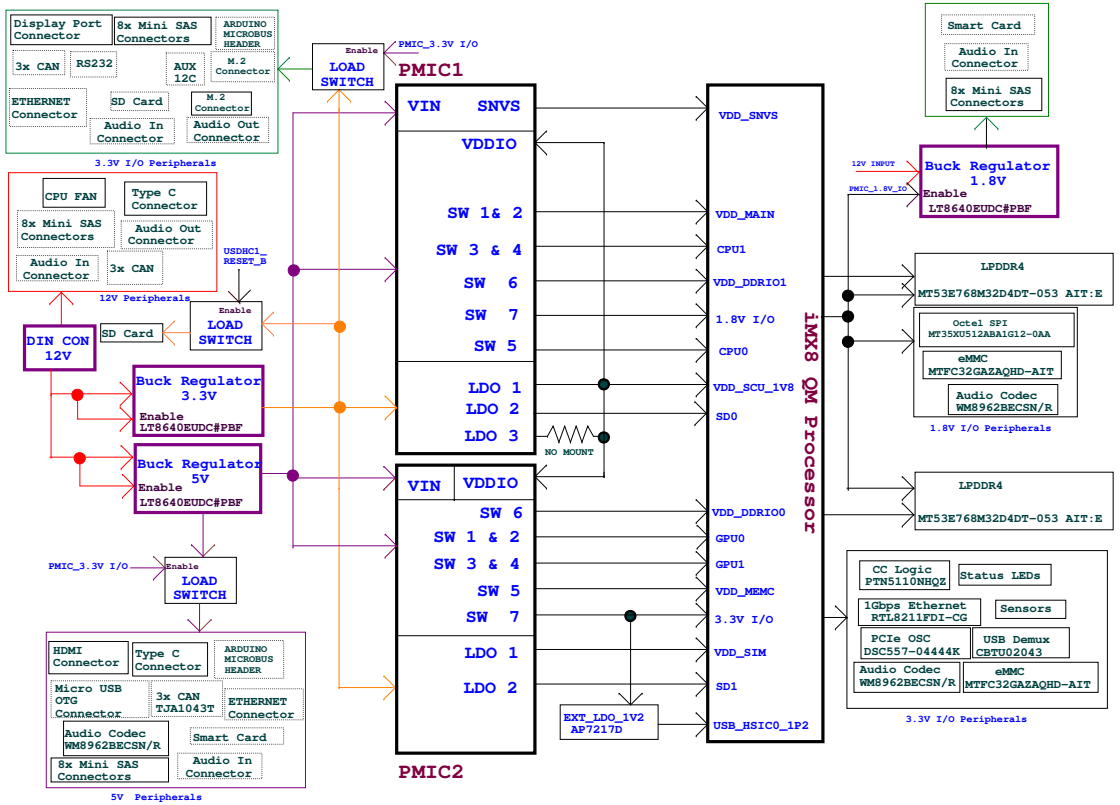
Date: Wednesday, September 03, 2025 | Sheet 2 of 32

i.MX 8QM CPU BOARD BLOCK_DIAGRAM



ICAP Classification:	CP: ___ IUO: ___ PUBI: X	
Drawing Title:	i.MX 8QM CPU CARD	
Page Title:	BLOCK DIAGRAM	
Size A2	Document Number SOURCE: SCH-29420, PDF: SPF-29420	Rev H1
Date: Wednesday, September 03, 2025	Sheet 3 of 32	

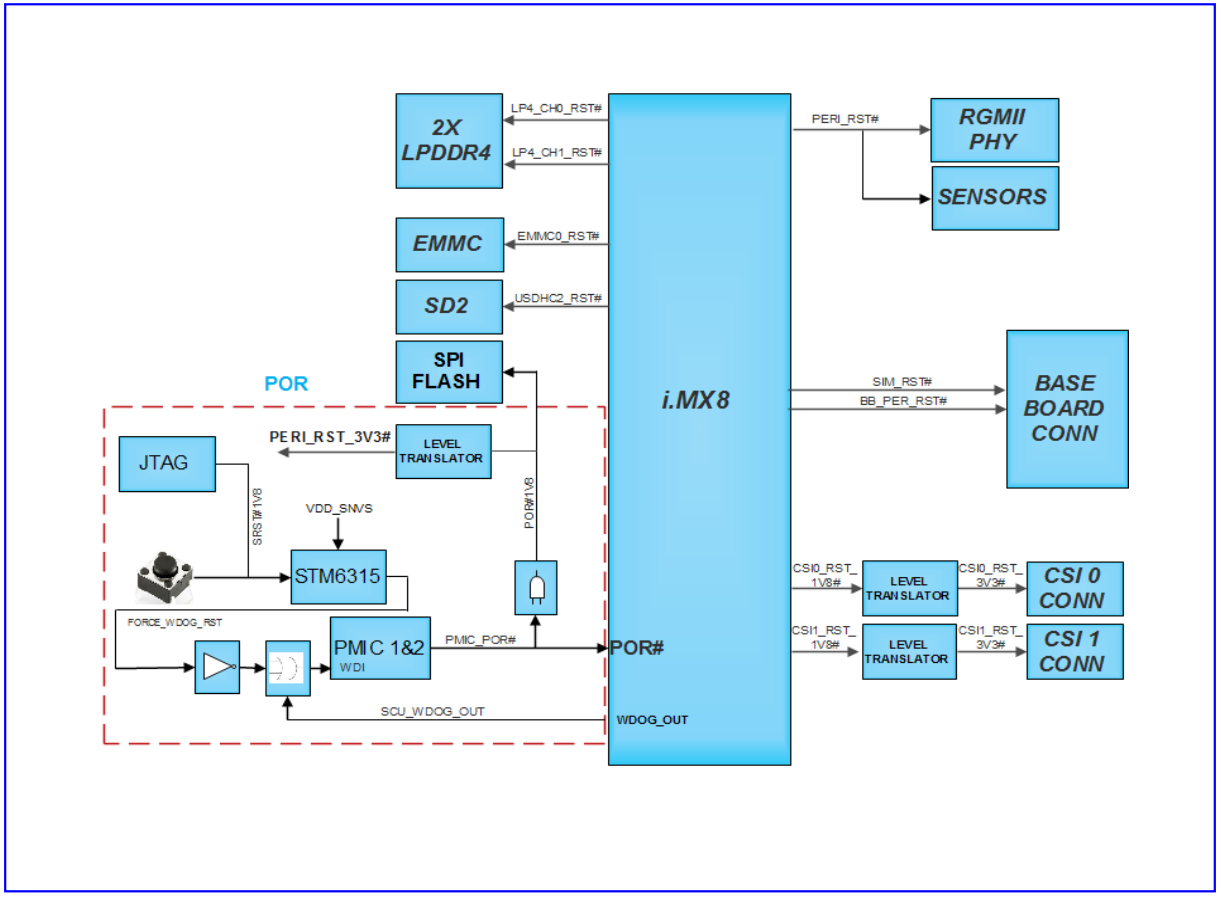
Power Flow Diagram



Note: Doted line represents components in Base Boad.

ICAP Classification:	CP: ___ IUO: ___ PUBI: _X_
Drawing Title:	i.MX 8QM CPU CARD
Page Title:	BLOCK DIAGRAM - POWER
Size: A3	Document Number: SOURCE: SCH-29420, PDF: SPF-29420 — Rev: H1
Date: Wednesday, September 03, 2025	Sheet 4 of 32

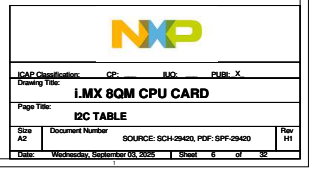
BLOCK DIAGRAM - RESET



ICAP Classification:	CP: ___ IUO: ___ PUB: X
Drawing Title:	i.MX 8QM CPU CARD
Page Title:	BLOCK DIAGRAM - RESET
Size: A2	Document Number: SOURCE: SCH-29420, PDF: SPF-29420 Rev: H1
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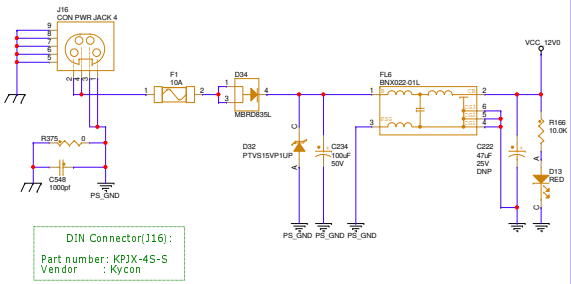
i.MX 8QM CPU CARD I2C TABLE

DEVICE	Location	Speed(kbps)	U Number	DEVICE ADDRESS	I2C	IO LEVEL
PMIC1	CPU	3400		0x08	PMIC I2C	1.8V
PMIC2	CPU	3400		0x09	PMIC I2C	1.8V
LSM303AGRTR	CPU	400	U21	0x19 & 0x1e*	I2C0	3.3V
MPL3115A2ST1	CPU	400	U14	0x60	I2C0	3.3V
I3G4250DTR	CPU	400	U13	0x69	I2C0	3.3V
PTN5110NHQZ	CPU	400	U37	0x51	I2C0	3.3V
ARDUINO/MIKROBUS	BASE				I2C0	3.3V
ENET CONN	BASE			0x68	I2C0	3.3V
AUX I2C	BASE			0x20	I2C0	3.3V
AUDIO IN/OUT	BASE			0x90	M41.I2C	1.8V
WM8962B	CPU			0x1A	I2C0	1.8V

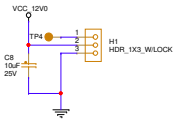


POWER SUPPLY

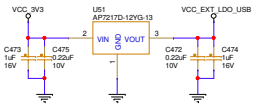
12V DC INPUT DIN CONNECTOR



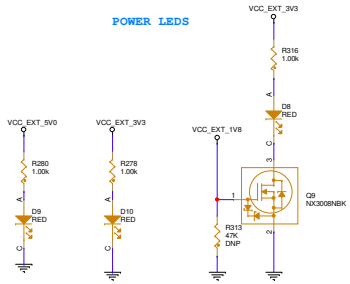
FAN CONNECTOR



LDO FOR 1V2:VDD_USB_HSIC0_1P2

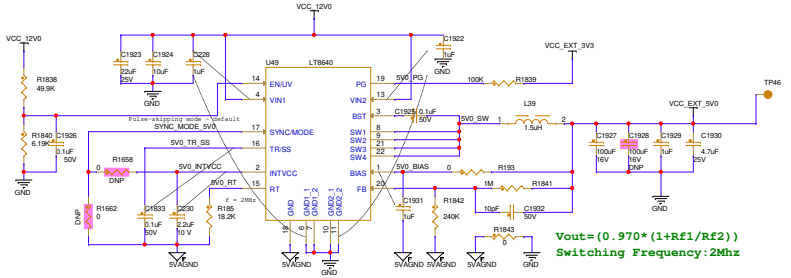


POWER LEDS



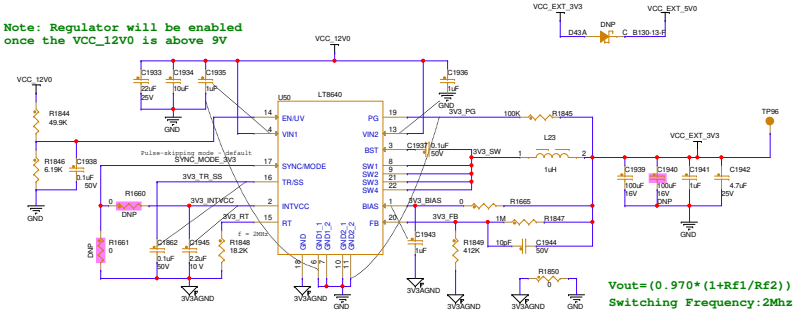
12V TO 5V

Note: Regulator will be enabled once the VCC_12V0 is above 9V



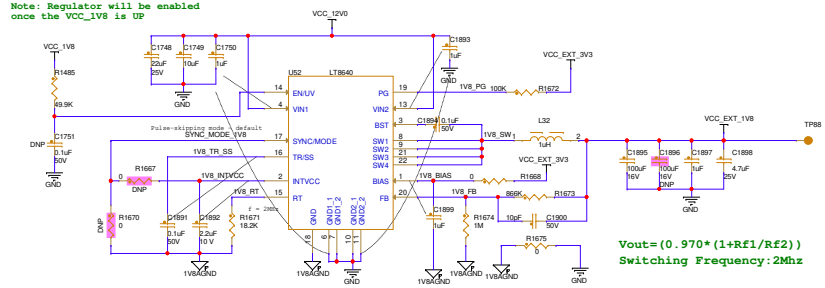
12V TO 3.3V

Note: Regulator will be enabled once the VCC_12V0 is above 9V



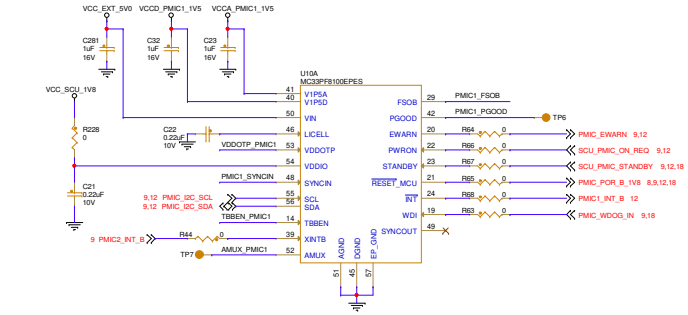
12V TO 1.8V

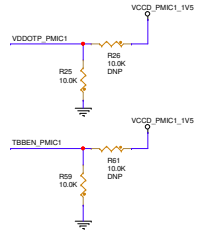
Note: Regulator will be enabled once the VCC_1V8 is UP

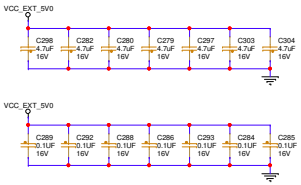


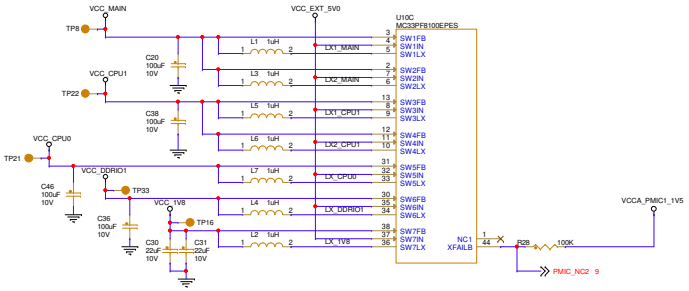
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Drawing Title:	i.MX 8QM CPU CARD		
Page Title:	POWER SUPPLY		
Size: A2	Document Number: SOURCE: SCH-29420, PDF: SPF-29420		Rev: H1
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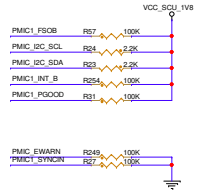
POWER SUPPLY - PMIC 1

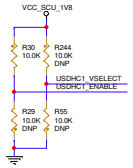


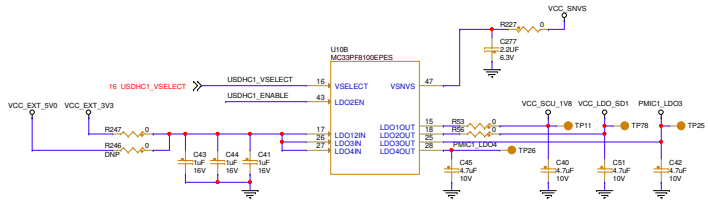


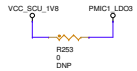




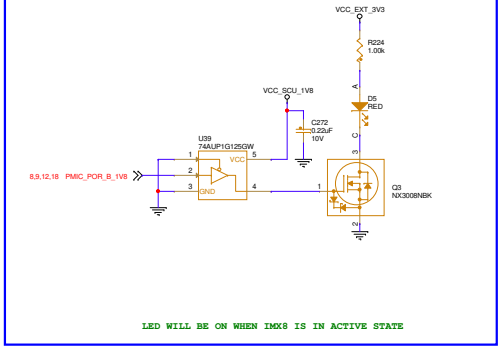








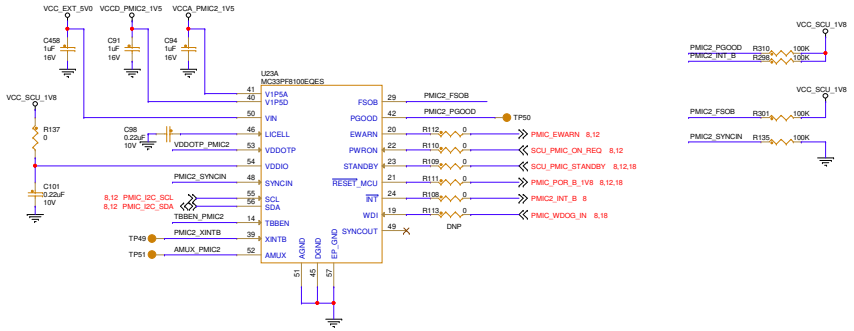
IMX8 RESET INDICATION

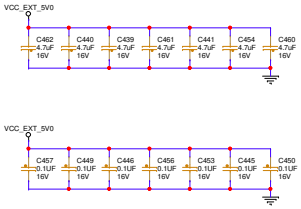


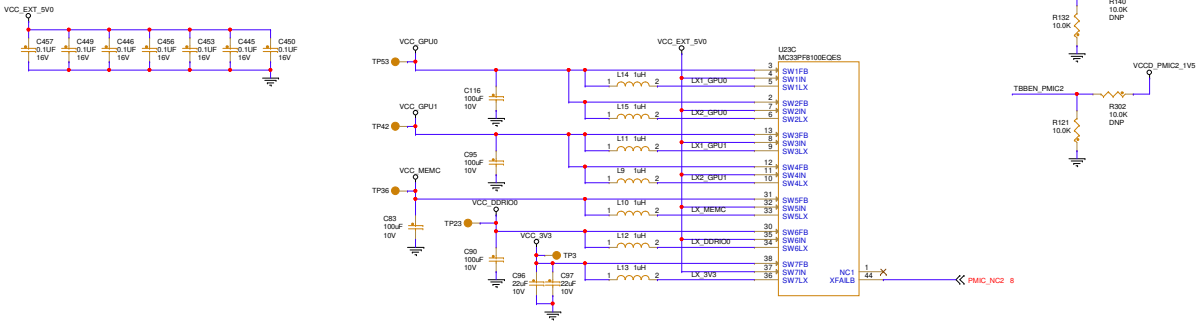
LED WILL BE ON WHEN IMX8 IS IN ACTIVE STATE

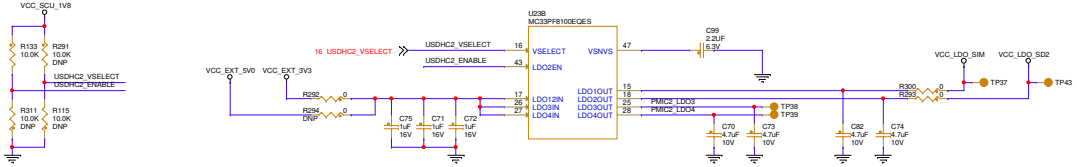
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POWER SUPPLY - PMIC 2

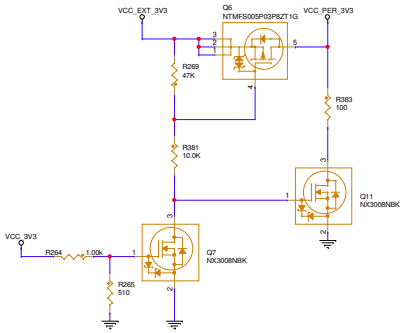




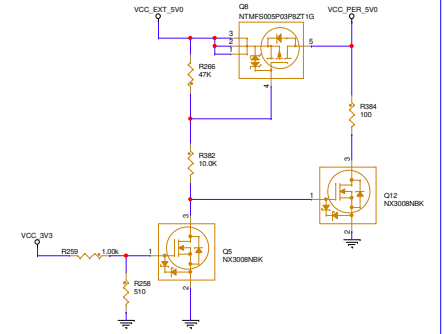




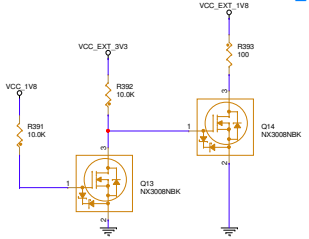
LOAD SWITCH FOR 3V3 PERIPHERALS



LOAD SWITCH FOR 5V0 PERIPHERALS



DISCHARGE CIRCUIT FOR EXT_1V8



ICAP Classification:	CP:	IUO:	PUBI: X
Drawing Title:	i.MX 8QM CPU CARD		
Page Title:	PMIC 2		
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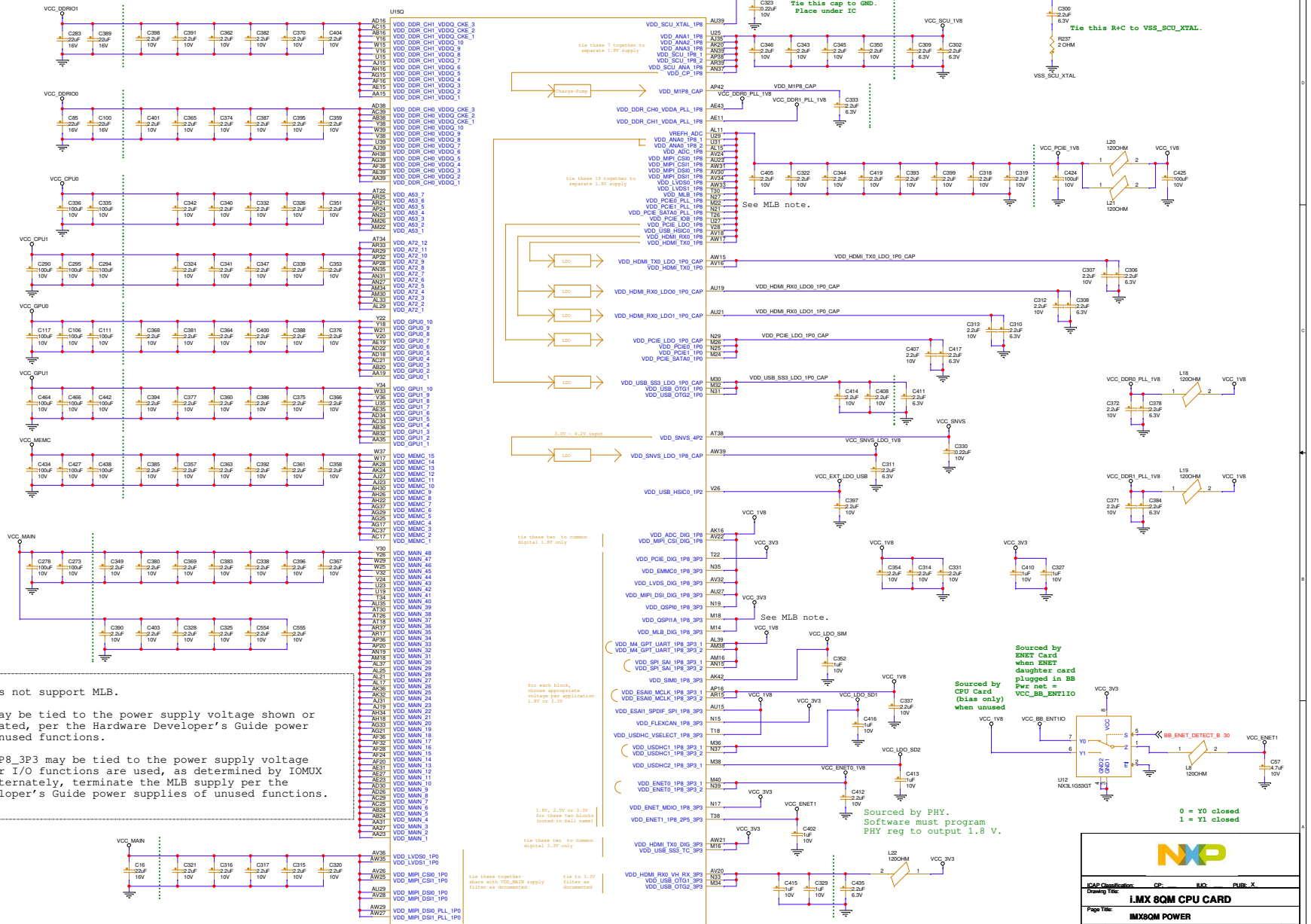
IMX8QM POWER

Mount devices between dashed green lines and U15 symbol under BGA.

Processor does not support MLB.

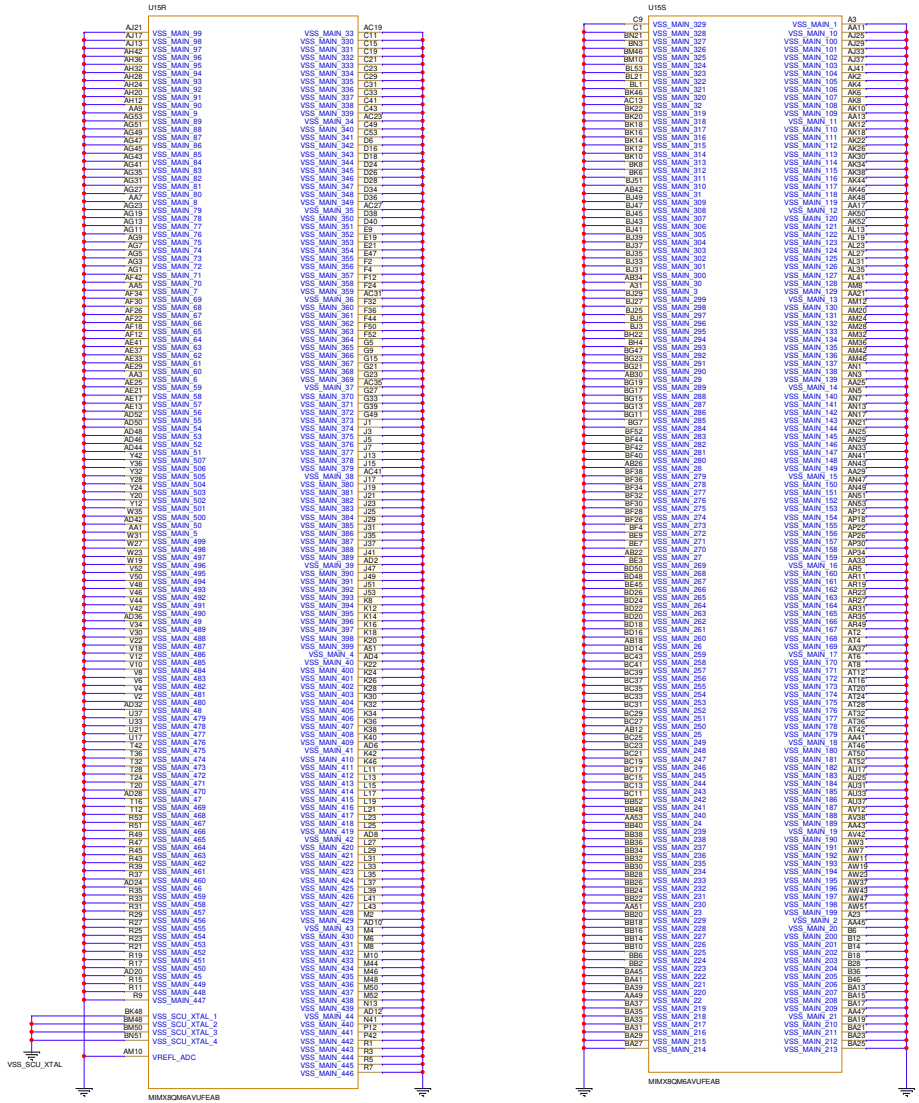
VDD_MLB_1P8 may be tied to the power supply voltage shown or may be terminated, per the Hardware Developer's Guide power supplies of unused functions.

VDD_MLB_DIG_1P8_3P3 may be tied to the power supply voltage shown if other I/O functions are used, as determined by IOMUX selection. Alternately, terminate the MLB supply per the Hardware Developer's Guide power supplies of unused functions.



ICAP Classification:	CP: ___ IUO: ___ PUB: X
Drawing Title:	i.MX 8QM CPU CARD
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IMX8QM GND SECTIONS



ICAP Classification: CP: ___ IUO: ___ PUBI: X

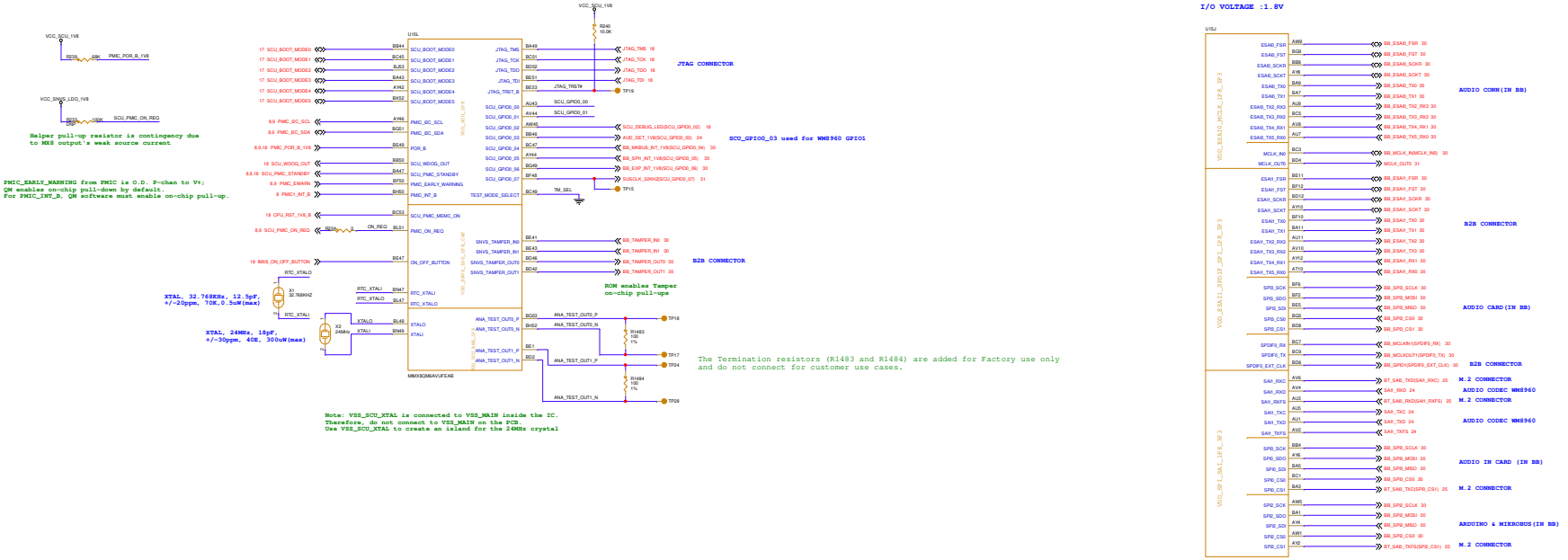
Drawing Title: i.MX 8QM CPU CARD

Page Title: IMX8QM GND

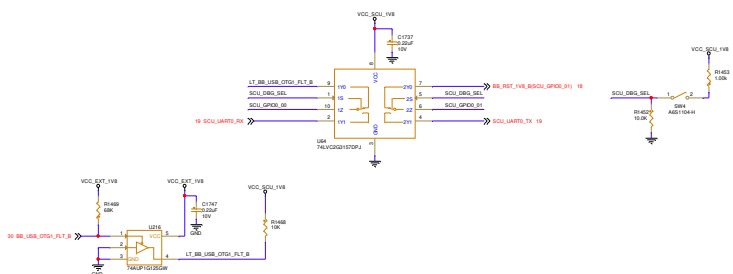
Size	Document Number	Rev
A2	SOURCE: SCH-29420, PDF: SPF-29420	H1

Date: Wednesday, September 03, 2025 Sheet 11 of 32

IMX8 SECTIONS_1



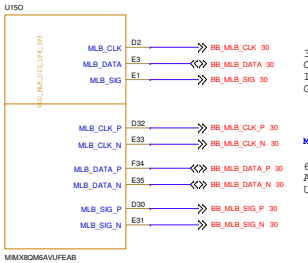
I/O VOLTAGE :1.8V



DAP Check/Section:	CP: A/D: PUB: X
Drawing Title:	i.MX 8QM CPU CARD
Page Title:	IMX8 SECTIONS 1
Size: A1	Document Number: SOURCE: SCH-29400, PDF: SPF-29400
Date:	Wednesday, September 04, 2019 Sheet 18 of 30

IMX8 SECTIONS_2

I/O VOLTAGE :3.3V

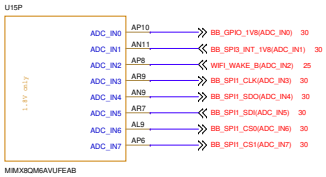


3-Wire Single-Ended MLB not supported by the processor.
Other functions are supported as determined by IOMUX selection.
If unused, adopters must follow the Hardware Developer's
Guide Unused input/output terminations section.

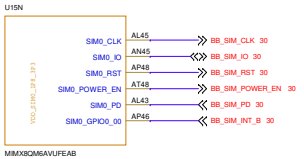
MLB(B2B CONNECTOR)

6-Wire Differential MLB not supported by the processor.
Adopters must follow the Hardware Developer's Guide
Unused input/output terminations section.

I/O VOLTAGE :1.8V

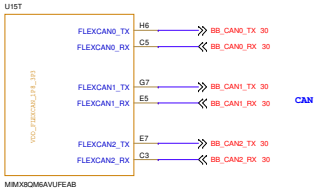


ENET CARD SPI(IN BB)



SIM(B2B CONNECTOR)

I/O VOLTAGE :3.3V

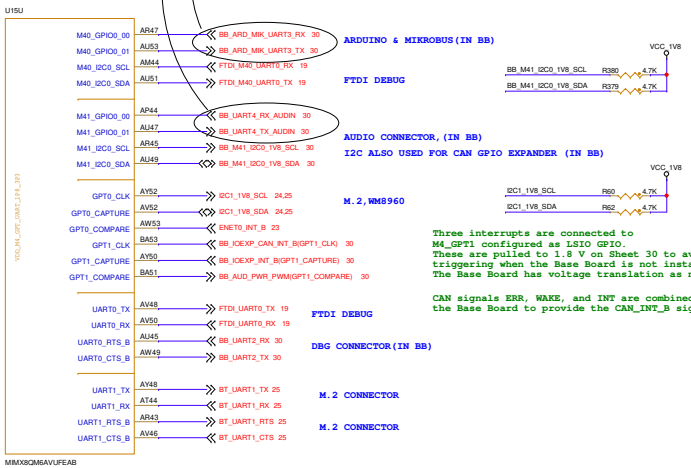


CAN 0,1,2(B2B CONNECTOR)

NXP is aware that the UART net names are inaccurate. There is no plan
to change the net names.

Adopters please note:
UART3 net name on contact AR47/AU53 should be UART4.
UART4 net name on contact AP44/AU47 should be UART3.
2-Wire AUDIN function not available on Base Board.

I/O VOLTAGE :1.8V



ARDUINO & MIKROBUS(IN BB)

FTDI DEBUG

AUDIO CONNECTOR,(IN BB)

I2C ALSO USED FOR CAN GPIO EXPANDER (IN BB)

M.2,WM8960

Three interrupts are connected to
M4_GPT1 configured as LSIO GPIO.
These are pulled to 1.8 V on Sheet 30 to avoid false
triggering when the Base Board is not installed.
The Base Board has voltage translation as needed.

CAN signals ERR, WAKE, and INT are combined together on
the Base Board to provide the CAN_INT_B signal.

FTDI DEBUG

DBG CONNECTOR(IN BB)

M.2 CONNECTOR

M.2 CONNECTOR

ICAP Classification: CP: ___ IUO: ___ PUBI: X

Drawing Title: **i.MX 8QM CPU CARD**

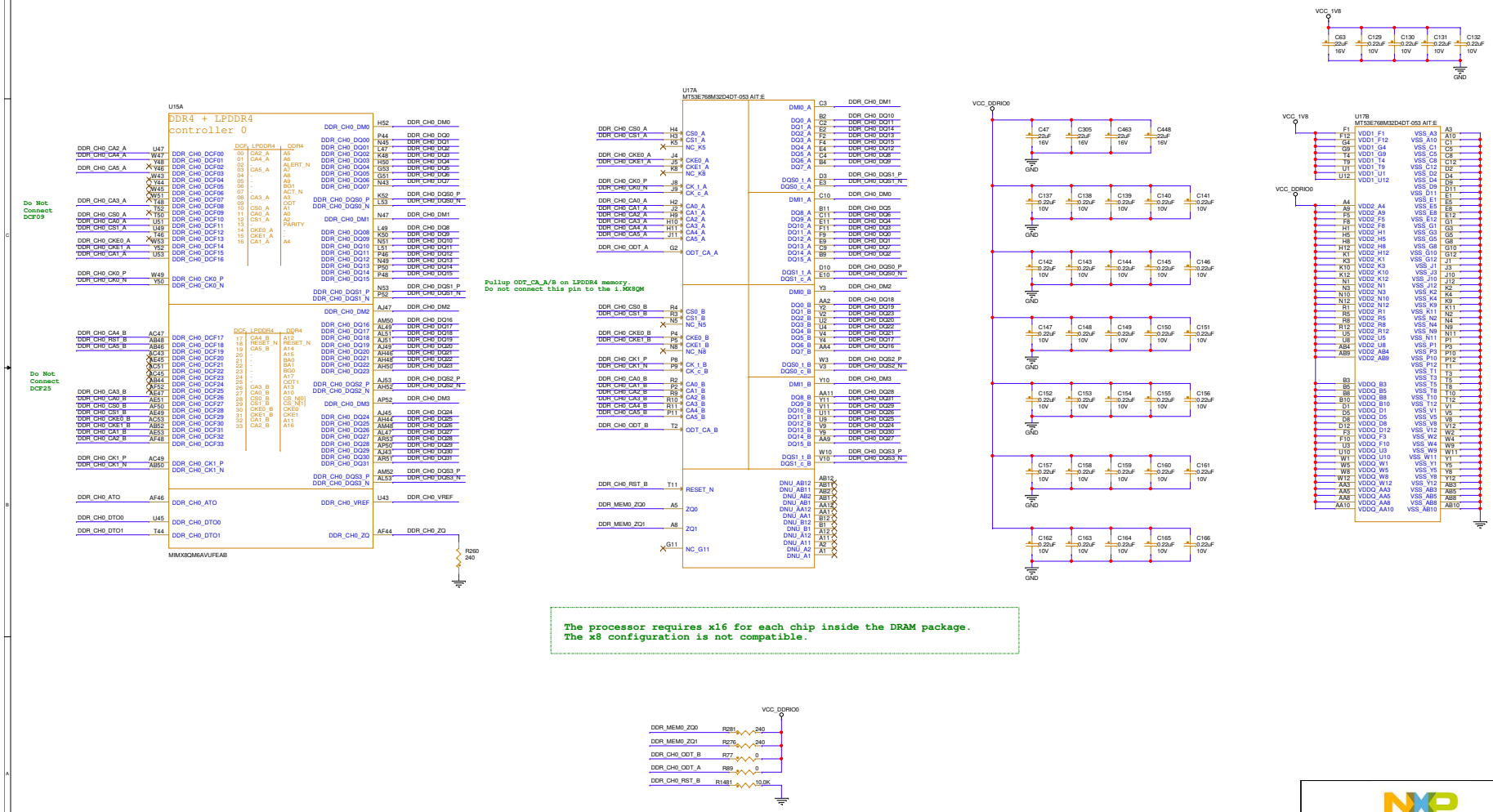
Page Title: **IMX8 SECTIONS 2**

Size	Document Number	Rev
A2	SOURCE: SCH-29420, PDF: SPF-29420	H1

Date: Wednesday, September 03, 2025 | Sheet 13 of 32

LPDDR4 DRAM 1 OF 2

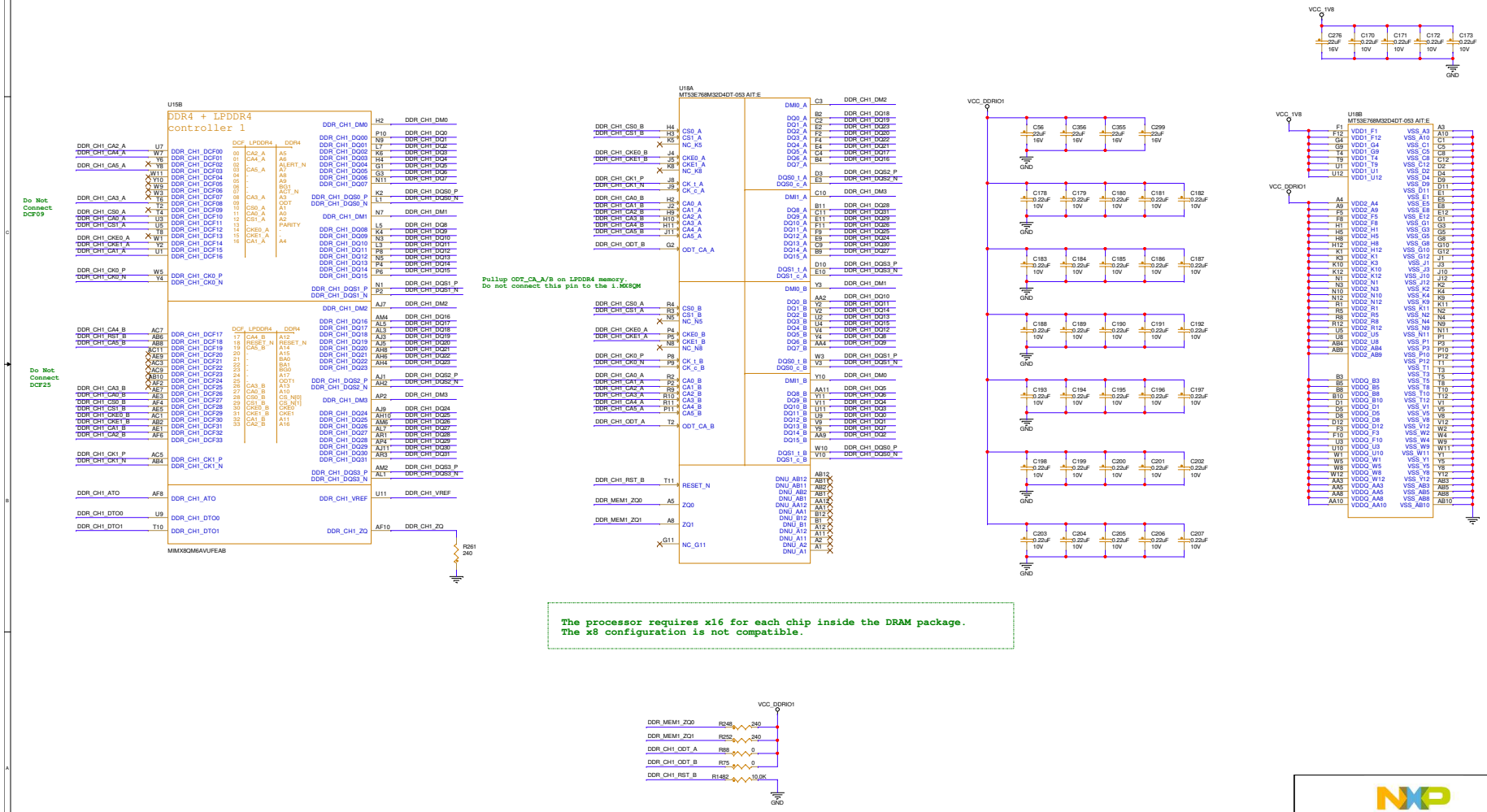
Total System DRAM = 6 Gbyte



ICAP Classification:	CP: ___ IUO: ___ PUBI: X
Drawing Title:	i.MX 8QM CPU CARD
Page Title:	LPDDR4 DRAM 1 OF 2
Size: A2	Document Number: SOURCE: SCH-29420, PDF: SPF-29420
Rev: H1	
Date: Wednesday, September 03, 2025	Sheet 14 of 32

LPDDR4 DRAM 2 OF 2

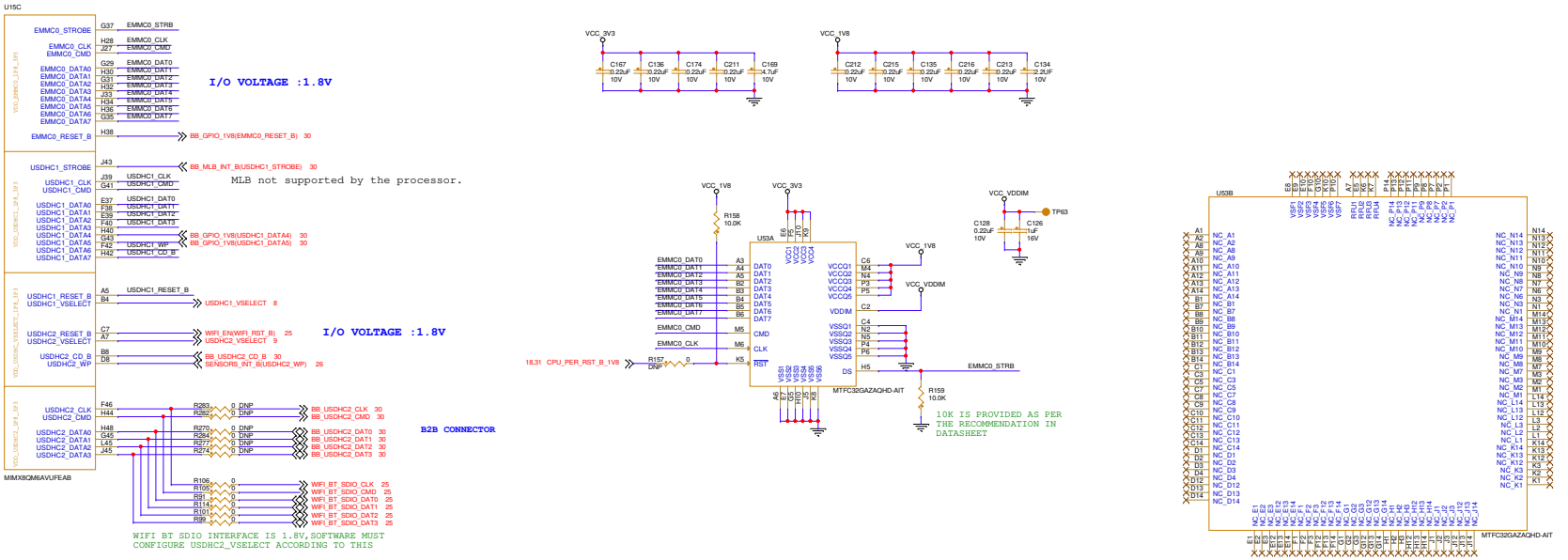
Total System DRAM = 6 Gbyte



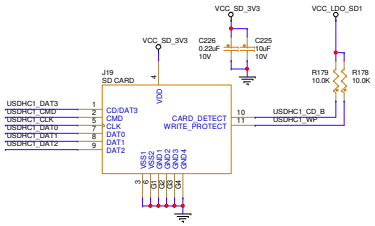
The processor requires x16 for each chip inside the DRAM package.
The x8 configuration is not compatible.

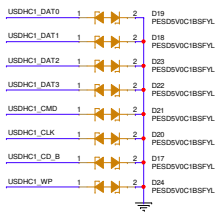
ICAP Classification:	CP: ___ IUO: ___ PUBI: X
Drawing Title:	i.MX 8QM CPU CARD
Page Title:	LPDDR4 DRAM 2 OF 2
Size: A2	Document Number: SOURCE: SCH-29420, PDF: SPF-29420 — Rev: H1
Date: Wednesday, September 03, 2025	Sheet 15 of 32

eMMC 32GB

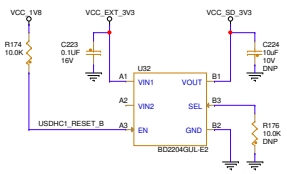


SD CARD INTERFACE





SDXC Power Control

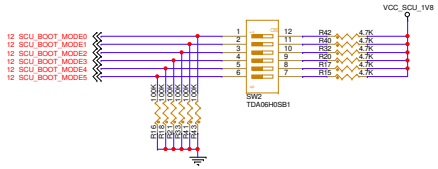


Bulk capacitance already present in this rail (C225)

SEL Input has Internal Pulldown(700k)

ICAP Classification:	CP: ___	IUO: ___	PUBI: X
Drawing Title:	i.MX 8QM CPU CARD		
Page Title:	eMMC & SD CARD		
Size: A2	Document Number: SOURCE: SCH-29420, PDF: SPF-29420		Rev: H1
Date: Wednesday, September 03, 2025	Sheet 16 of 32		

BOOT CONFIGURATIONS



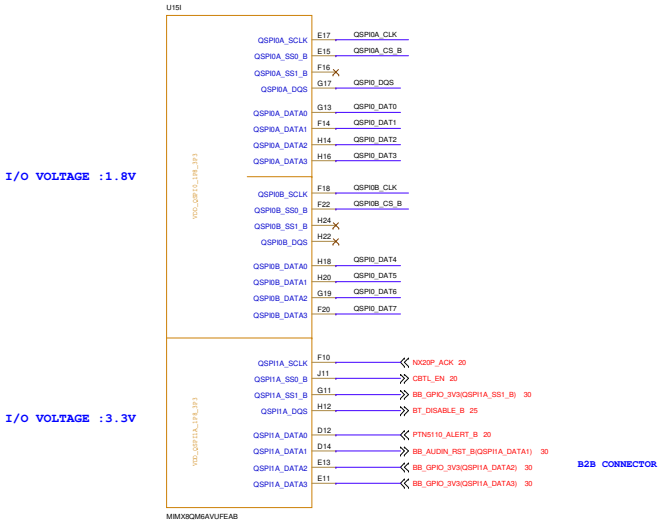
BOOT MODE

MODE	5	4	3	2	1	0
FUSE	0	0	0	0	0	0
SERIAL BOOT	0	0	0	1	0	0
eMMC0	0	0	1	0	0	0
SD1	0	0	1	1	0	0
Octal SPI	0	1	1	0	0	0

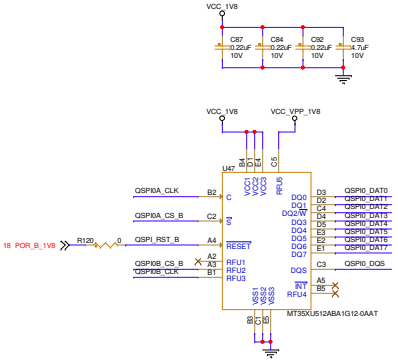
OCTAL/XSPI/QSPI FLASH

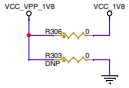
I/O VOLTAGE :1.8V

I/O VOLTAGE :3.3V

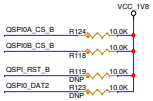


B2B CONNECTOR



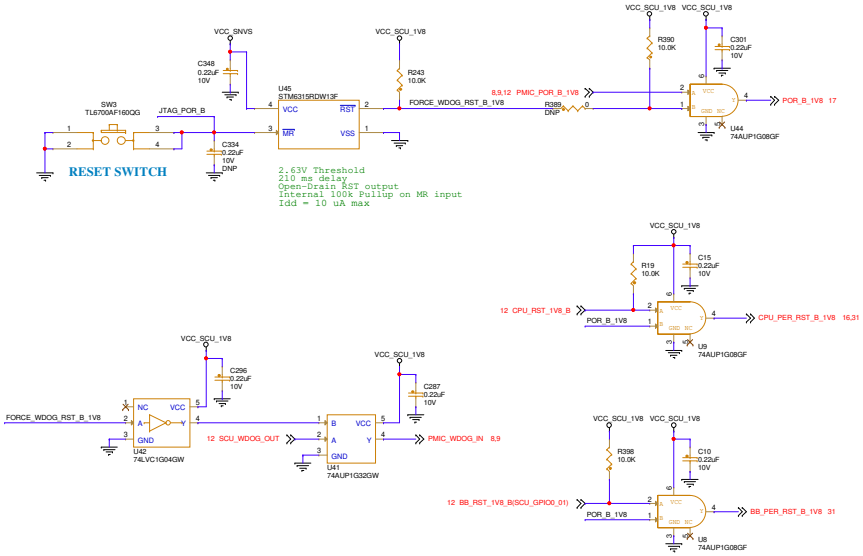


Use Pad sharing for these 2 RES

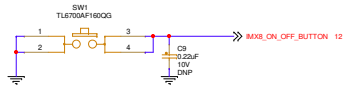


ICAP Classification:	CP: ___	IUO: ___	PUBI: X
Drawing Title:	i.MX 8QM CPU CARD		
Page Title:	BOOT CONFIGURATION & SPI FLASH		
Size A2	Document Number SOURCE: SCH-29420, PDF: SPF-29420		Rev H1
Date: Wednesday, September 03, 2025	Sheet 17 of 32		

RESET GENERATION

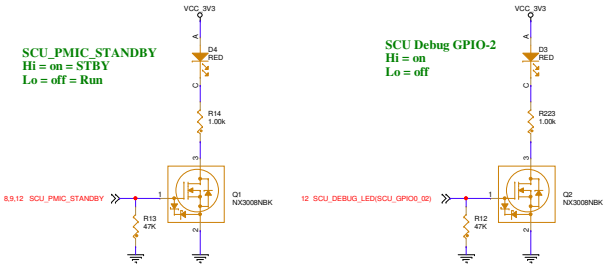


SYSTEM ON/OFF

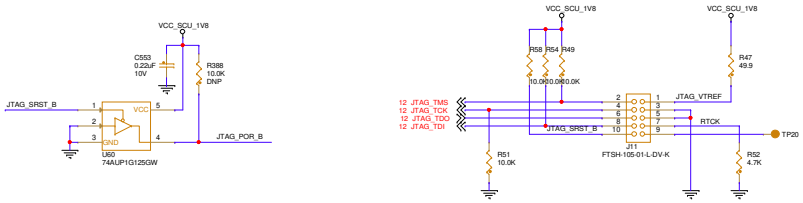


* Hold SW1 for 5 sec for force OFF
* Hold SW1 for 0.5 sec to turn ON

LED INDICATIONS



JTAG

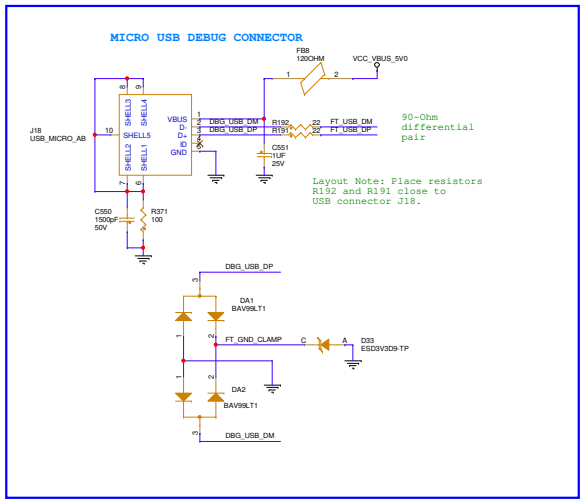


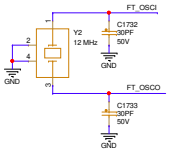
MX8QM On-Chip 50 kohm Pulls

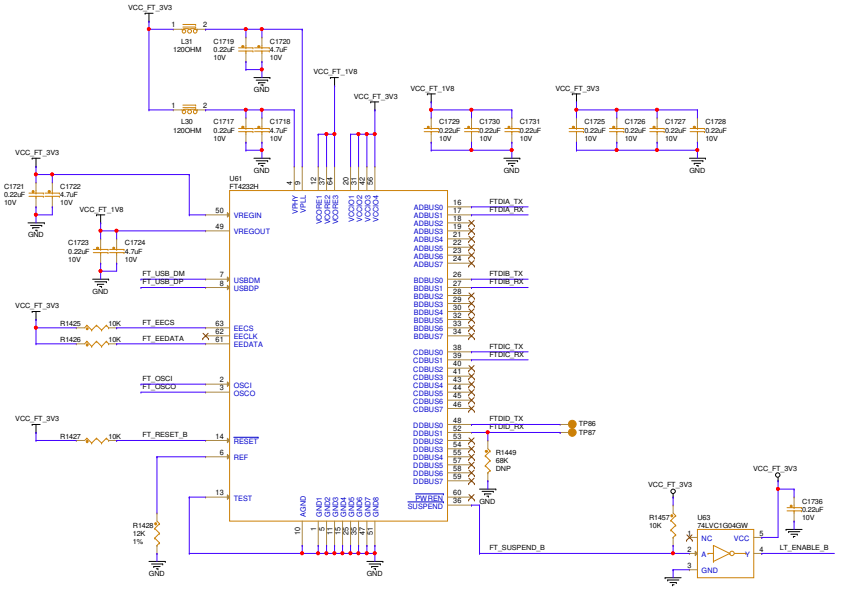
JTAG_TMS = PU
JTAG_TCK = PD
JTAG_TDI = PU
JTAG_TRST_B = PU
TEST_MODE_SELECT = PD

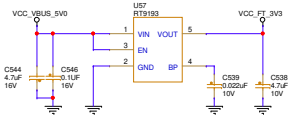
ICAP Classification: CP: ___ IUO: ___ PUBI: X
Drawing Title: i.MX 8QM CPU CARD
Page Title: RESET LOGIC & JTAG
Size A2 | Document Number SOURCE: SCH-29420, PDF: SPF-29420 | Rev H1
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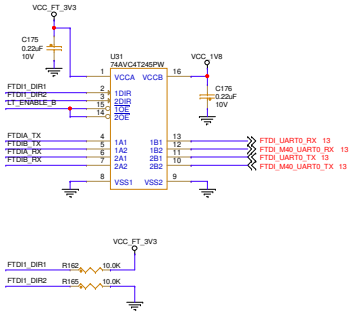
DEBUG UART-USB

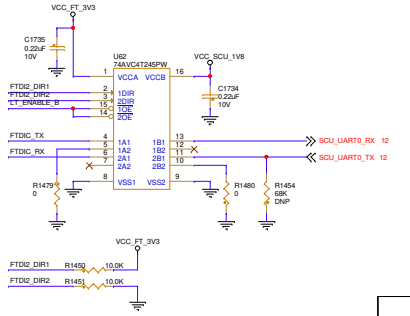


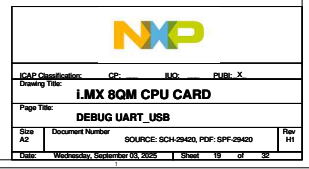






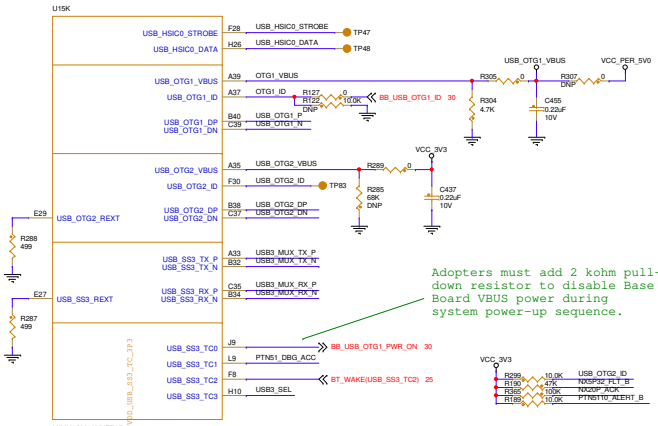






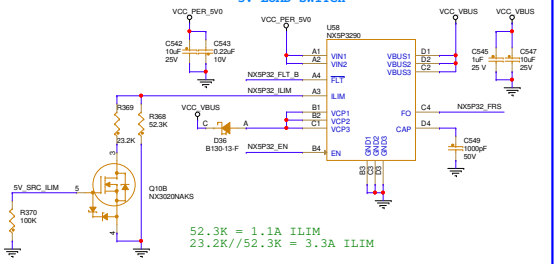
USB 3.x Type C

I/O VOLTAGE :3.3V



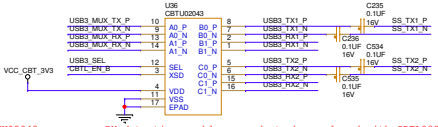
Adopters must add 2 kohm pull-down resistor to disable Base Board VBUS power during system power-up sequence.

5V LOAD SWITCH

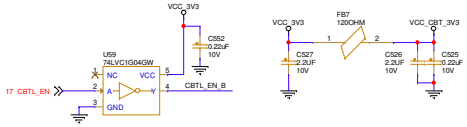


52.3K = 1.1A ILIM
23.2K//52.3K = 3.3A ILIM

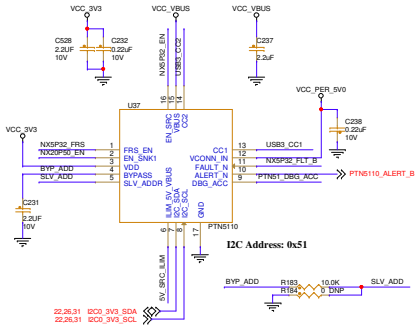
DIFFERENTIAL CHANNEL CROSSBAR SWITCH



NOTE: CBTU02043 may cause RX detection problem, needs to be replaced with CBTL02043.



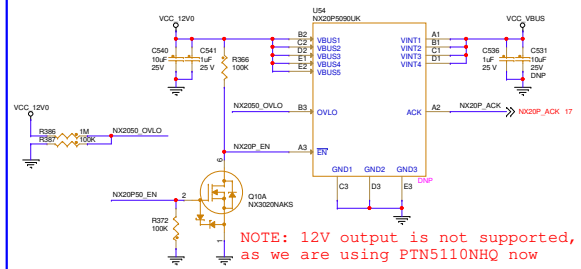
CC LOGIC DETECTION



I2C ADDRESS SELECTION

SLV_ADDR PIN	ADDRESS
GND	1010000x
100K PULL UP TO BYPASS	1010011x
UNCONNECTED	1010010x
10K PULL UP TO BYPASS	1010001x

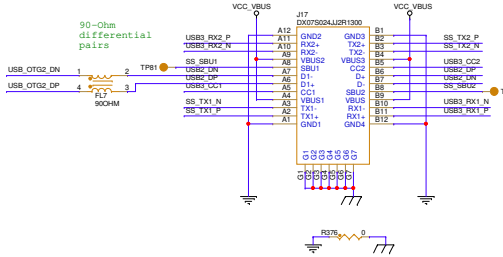
12V LOAD SWITCH



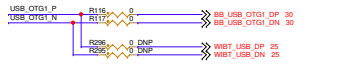
NOTE: 12V output is not supported, as we are using PTN5110NHQ now

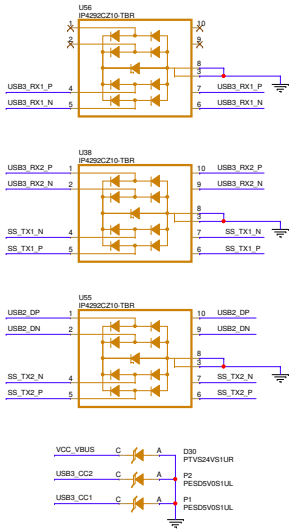
USB3.0 TYPE C CONNECTOR

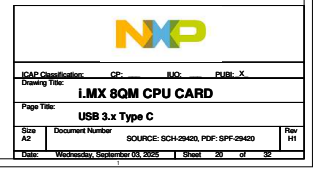
5V,3.3A
12V,3A



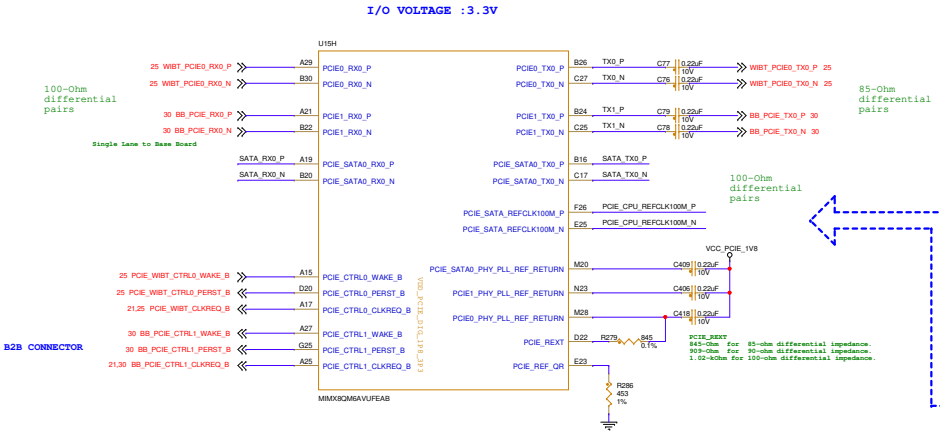
OTG1 Can be connected to M.2 Connector or Baseboard USB2.0 port using these resistor options.

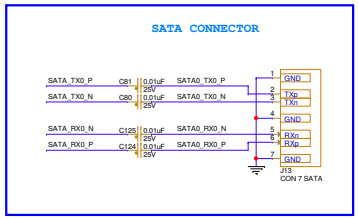






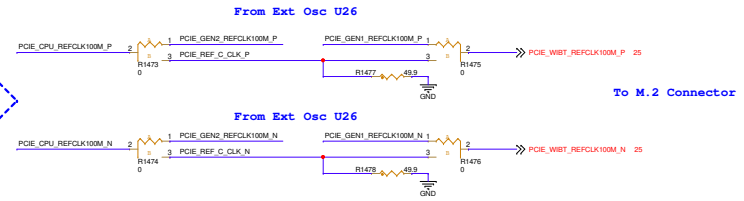
PCIe & SATA

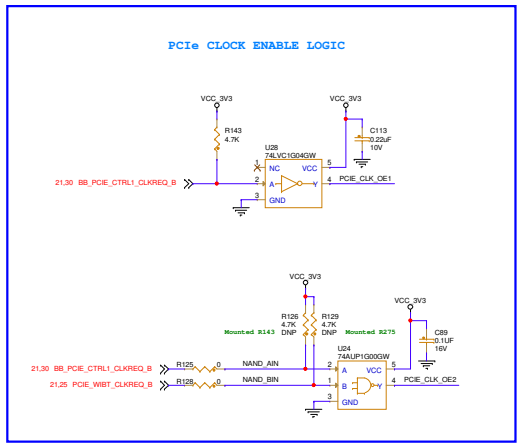


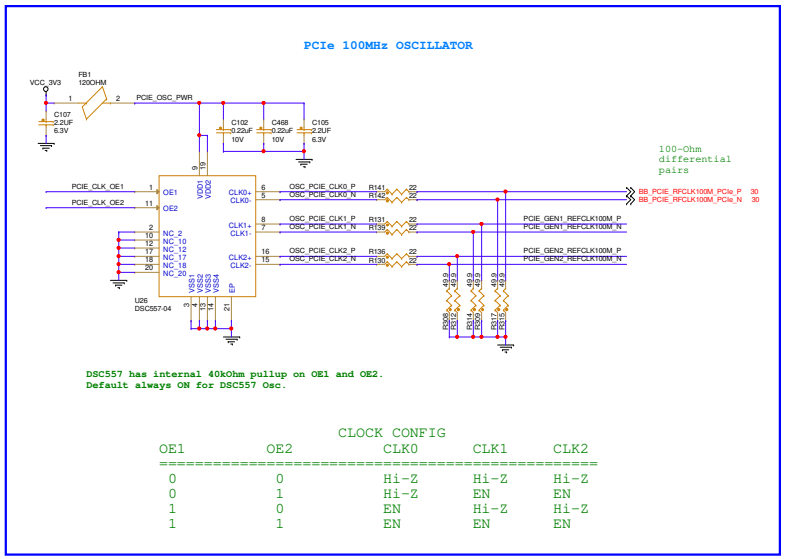


M.2 Connector on CPU Card has two PCIe Clock options:
(Set resistor strapping as per table)
1. Processor (NXP experimental use only; not recommended for customer use)
2. External Clock generator (By default)

M.2 Connector on Base Card has PCIe Clock
Option only from External clock generator.
(Set resistor strapping as per table)







CLOCK CONFIG

OE1	OE2	CLK0	CLK1	CLK2
0	0	Hi-Z	Hi-Z	Hi-Z
0	1	Hi-Z	EN	EN
1	0	EN	Hi-Z	Hi-Z
1	1	EN	EN	EN

	Option Resistors	Clock Source	
		Processor	Clock Generator
M.2 Connector on CPU Card	R1473	POS B	POS A
	R1474	POS B	POS A
	R1475	POS B	POS A
	R1476	POS B	POS A
	R128	DNP	MOUNT
	R129	MOUNT	DNP
	R130	DNP	MOUNT
	R131	DNP	MOUNT
	R136	DNP	MOUNT
	R139	DNP	MOUNT
M.2 Connector on Base Card	R1473	NA	POS A
	R1474	NA	POS A
	R1475	NA	POS A
	R1476	NA	POS A
	R128	NA	MOUNT
	R129	NA	DNP
	R130	NA	MOUNT
	R131	NA	MOUNT
	R136	NA	MOUNT
	R139	NA	MOUNT

ICAP Classification: CP: ___ IUO: ___ PUBI: X

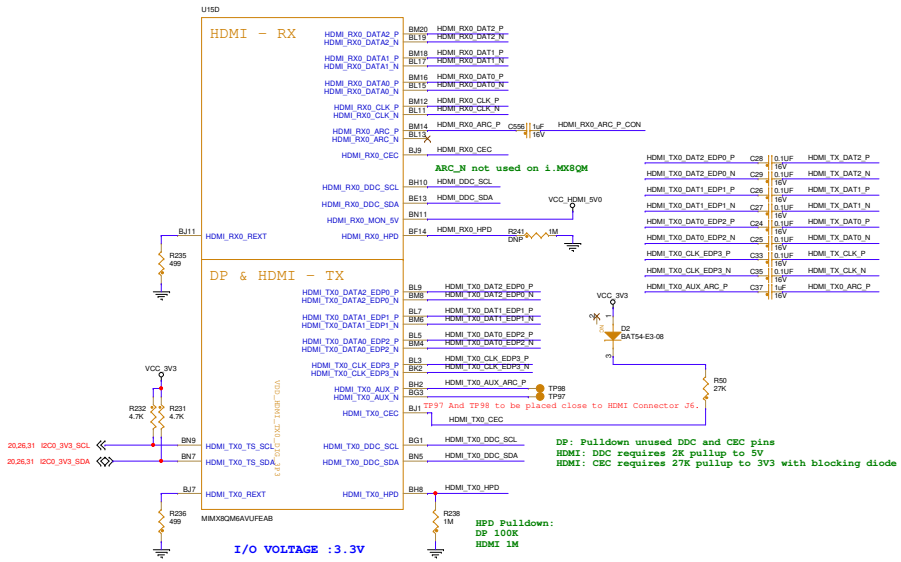
Drawing Title: i.MX 8QM CPU CARD

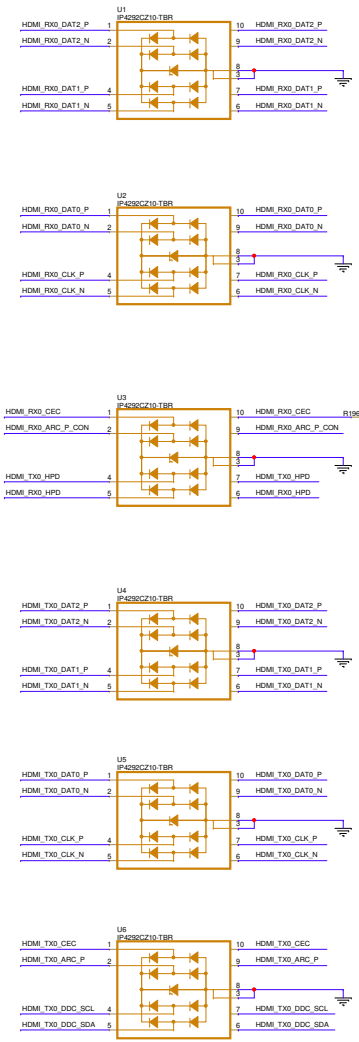
Page Title: PCIe & SATA

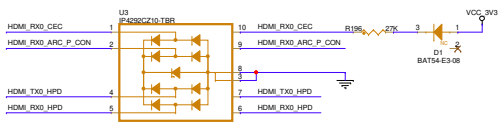
Size A2 | Document Number: SOURCE: SCH-29420, PDF: SPF-29420 | Rev H1

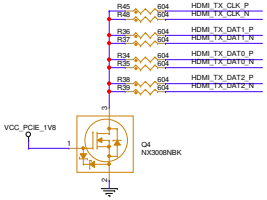
Date: Wednesday, September 03, 2025 | Sheet 21 of 32

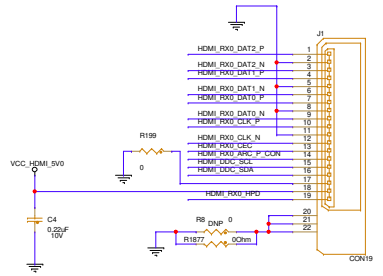
HDMI TX & RX

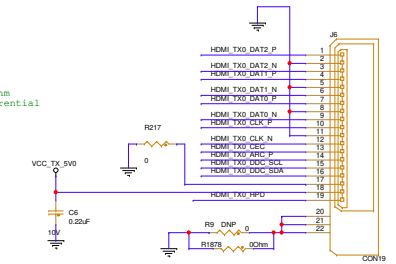








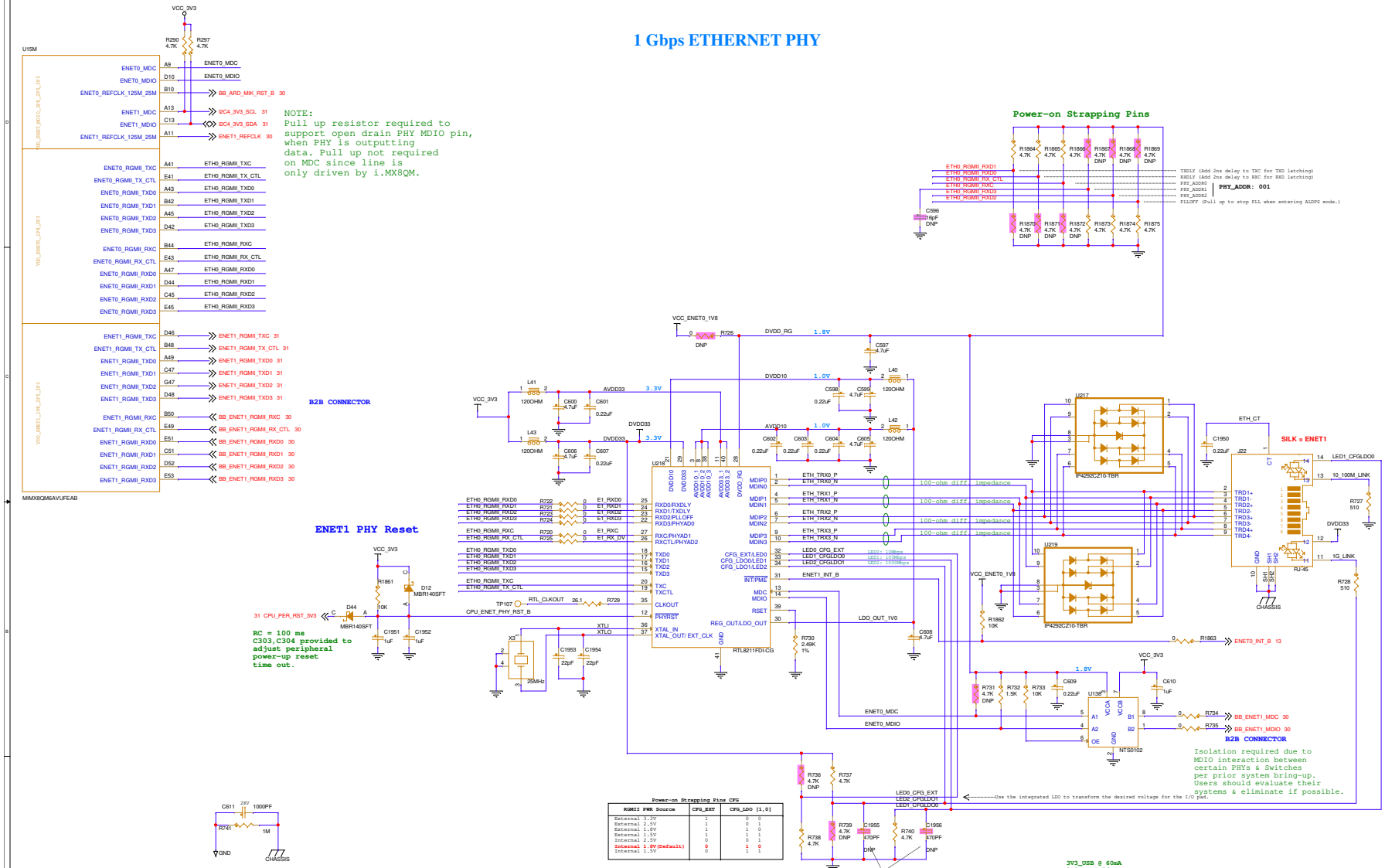




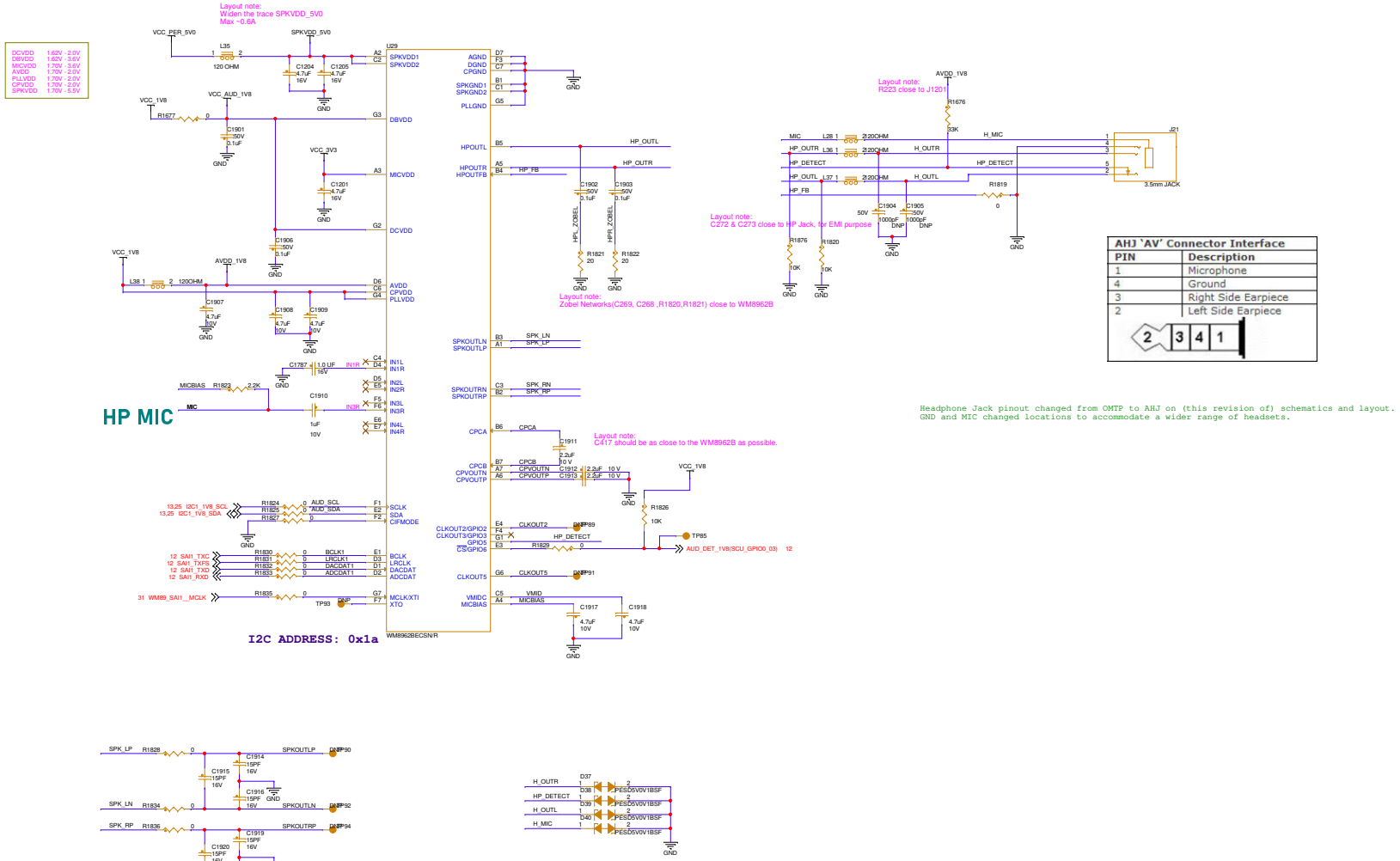
100-Ohm differential pairs

ICAP Classification:	CP: ___ IUO: ___ PUB: X
Drawing Title:	i.MX 8QM CPU CARD
Page Title:	HDMI TX & RX
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1 Gbps ETHERNET PHY



AUDIO CODEC WM8962

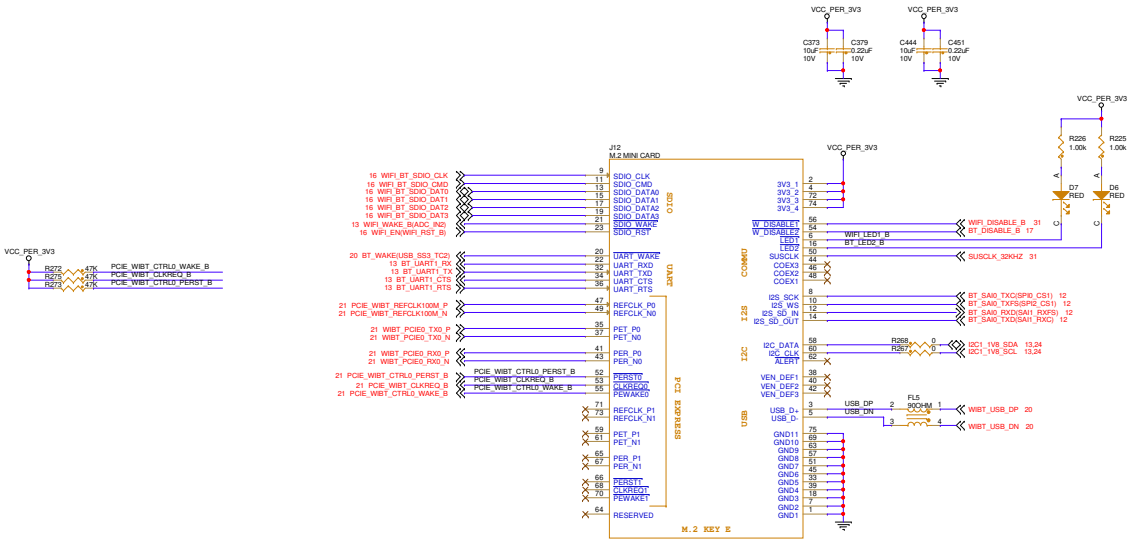


AHJ 'AV' Connector Interface	
PIN	Description
1	Microphone
4	Ground
3	Right Side Earpiece
2	Left Side Earpiece

Headphone Jack pinout changed from OMTP to AHJ on (this revision of) schematics and layout.
GND and MIC changed locations to accommodate a wider range of headsets.

ICAP Classification:	CP: ___ IUO: ___ PUB: X
Drawing Title:	i.MX 8QM CPU CARD
Page Title:	AUDIO CODEC WM8962
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WIFI_BLUETOOTH -M.2 CONNECTOR E-KEY



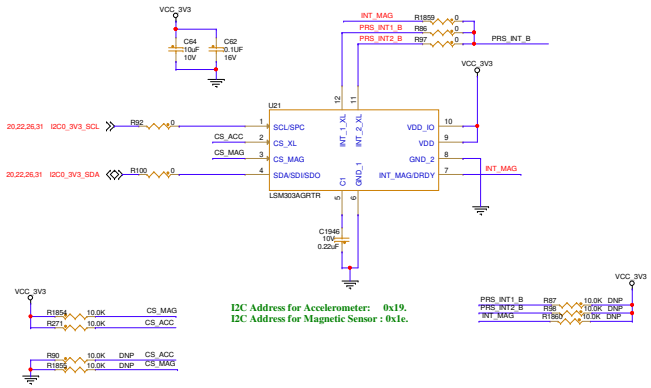
Add On Card used will be M.2 with E-Key Type 30x30 Dimension.

Information on compatible cards is provided on the nxp.com website.

ICAP Classification:	CP: ___ IUO: ___ PUBI: X
Drawing Title:	i.MX 8QM CPU CARD
Page Title:	WIFI_BLUETOOTH -M.2 CONNECTOR E-KEY
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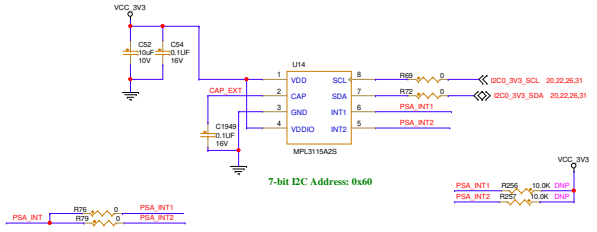
SENSORS

ACCELEROMETER



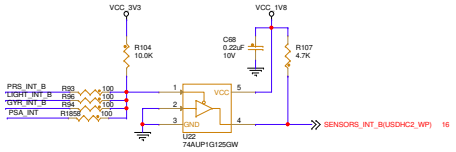
I2C Address for Accelerometer: 0x19.
I2C Address for Magnetic Sensor : 0x1e.

PRESSURE SENSOR WITH ALTIMETRY



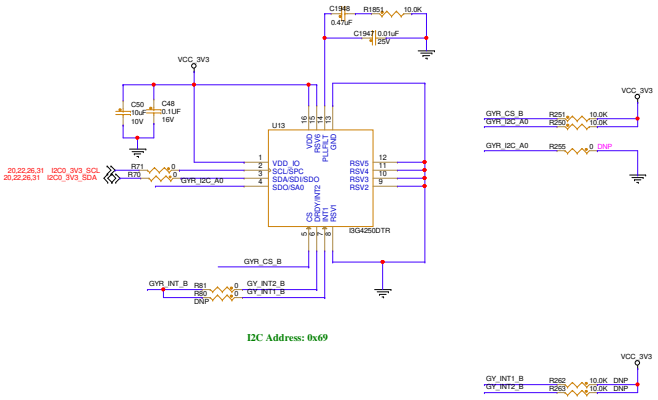
7-bit I2C Address: 0x60

SENSOR INTERRUPT



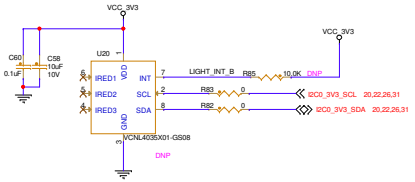
CONFIGURE ACCELEROMETER,GYROSCOPE,PRESSURE SENSOR
INTERRUPTS AS OPEN DRAIN,ACTIVE LOW IN RESPECTIVE REGISTERS

GYROSCOPE



I2C Address: 0x69

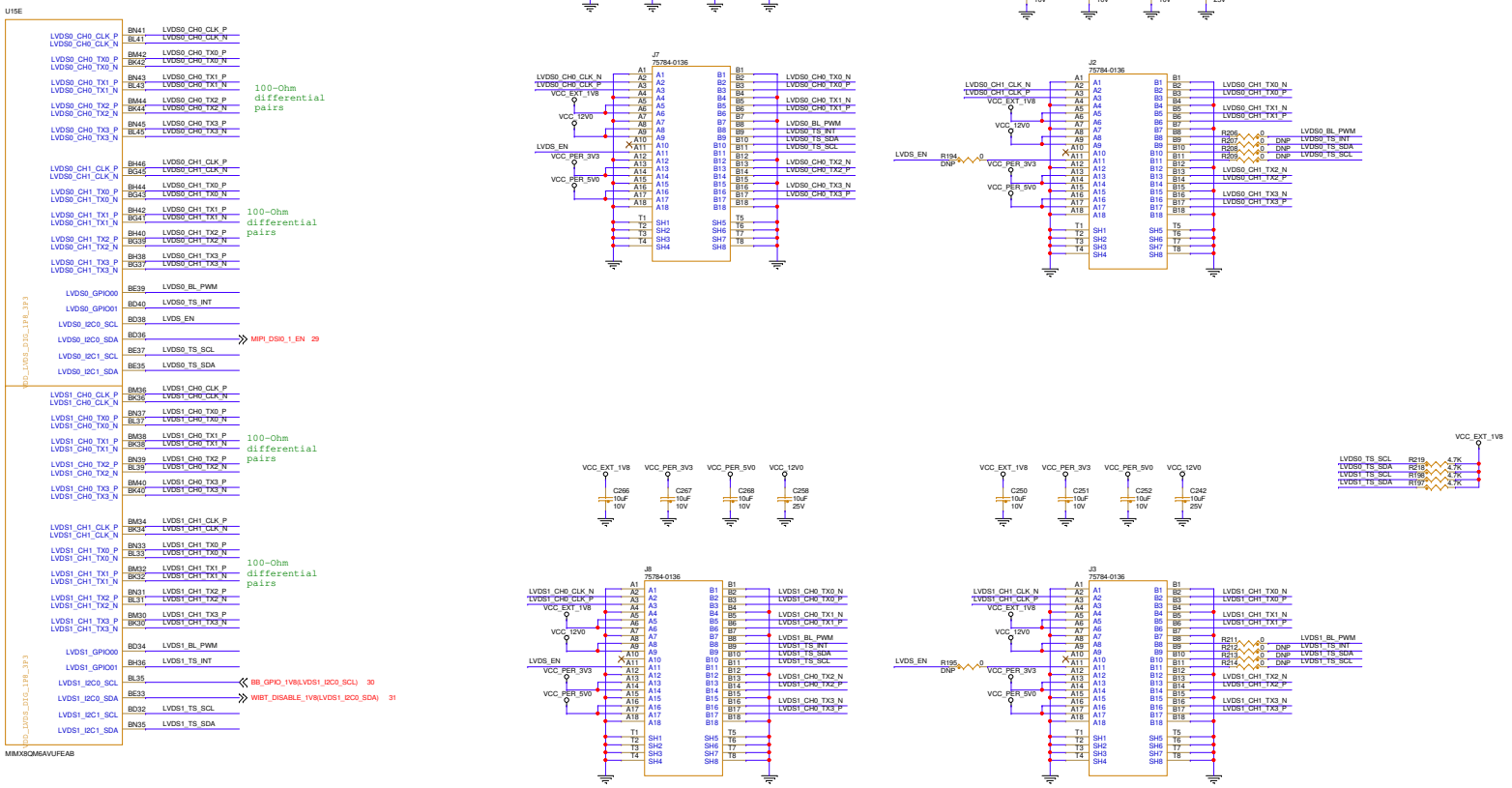
AMBIENT LIGHT SENSOR



7-bit I2C Address: 0x51

ICAP Classification:	CP: ___ IUO: ___ PUBI: X
Drawing Title:	i.MX 8QM CPU CARD
Page Title:	SENSORS
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LVDS0 CH0 & CH1 CONNECTORS



NXP

ICAP Classification: CP: ___ IUO: ___ PUBI: X

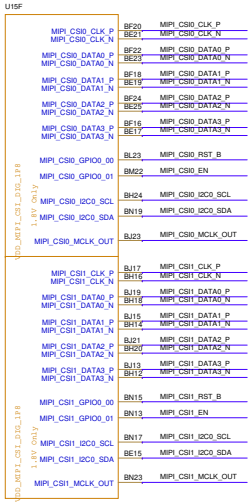
Drawing Title: i.MX 8QM CPU CARD

Page Title: LVDS0 CONNECTORS

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A2	SOURCE: SCH-29420, PDF: SPF-29420	H1

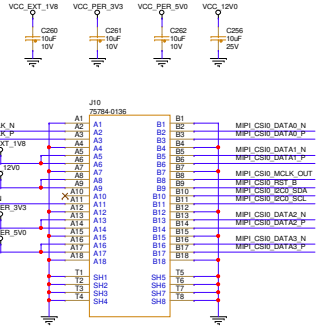
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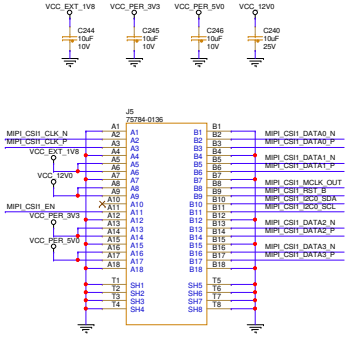
MIPI CSI CONNECTORS

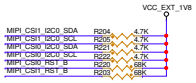


100-Ohm differential pairs

100-Ohm differential pairs

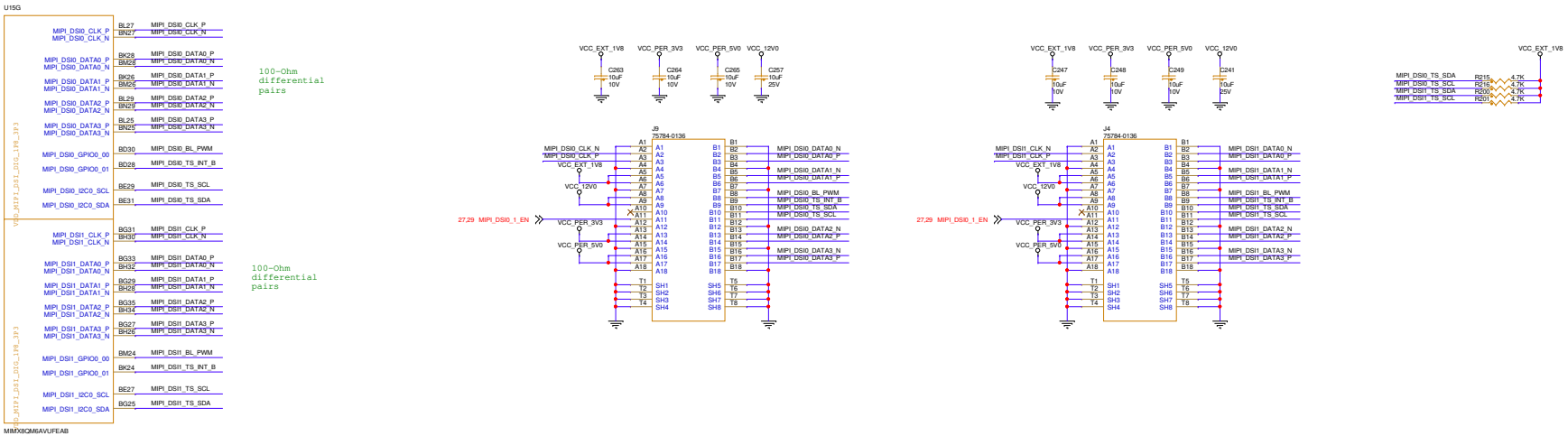






ICAP Classification:	CP: ___	IUO: ___ PUBI: X
Drawing Title:	i.MX 8QM CPU CARD	
Page Title:	MIPI CSI CONNECTORS	
Size A2	Document Number SOURCE: SCH-29420, PDF: SPF-29420	Rev H1
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MIPI DSI CONNECTORS



ICAP Classification: CP: ___ IUO: ___ PUBI: X

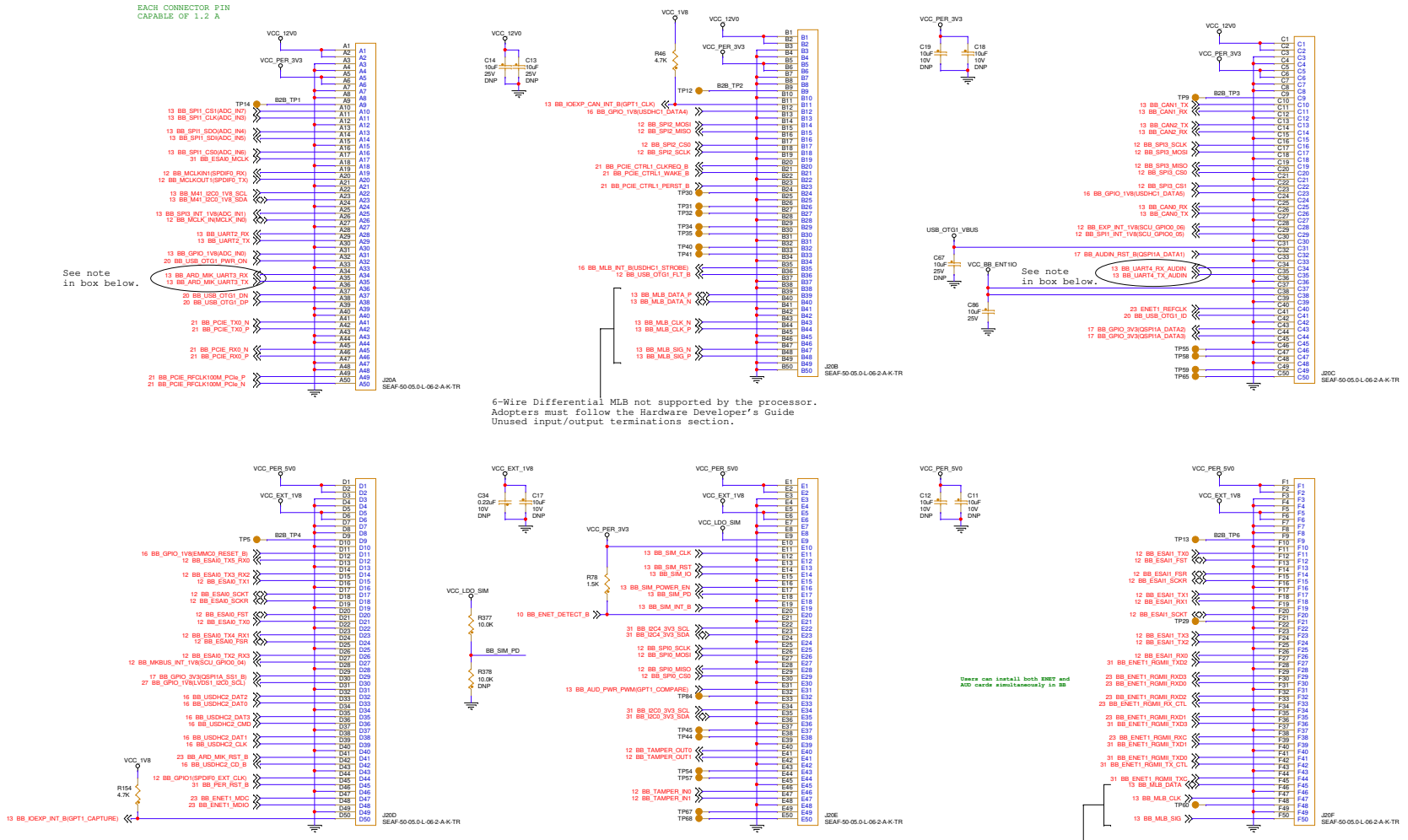
Drawing Title: i.MX 8QM CPU CARD

Page Title: MIPI DSI CONNECTORS

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B2B CONNECTOR



Users can install both ENET and AUD cards simultaneously in BB

NXP is aware that the UART net names circled are inaccurate. There is no plan to change the net names.

Adopters please note:
UART3 net name on pins A34 & A35 should be UART4.
UART4 net name on pins C34 & C35 should be UART3.
2-Wire AUDIN function not available on Base Board.

SOME PINS ARE RESERVED TO FACILITATE A COMMON BASE BOARD FOR QM AND QXP.

3-Wire Single-Ended MLB not supported by the processor.
Other functions are supported as determined by IOMUX selection.
If unused, adopters must follow the Hardware Developer's Guide Unused input/output terminations section.

ICAP Classification: CP: ___ IUO: ___ PUBI: X

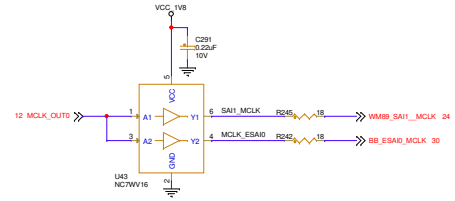
Drawing Title: i.MX 8QM CPU CARD

Page Title: B2B CONNECTOR

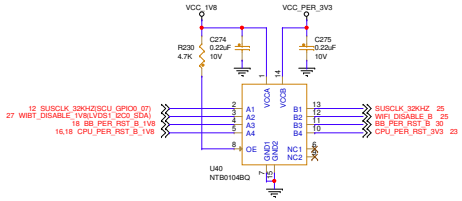
Size	Document Number	Rev
A2	SOURCE: SCH-29420, PDF: SPF-29420	H1

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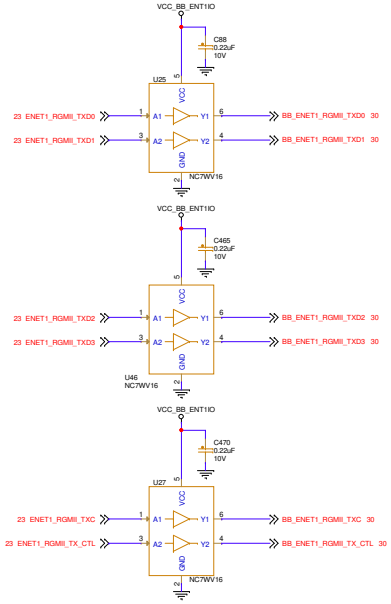
MCLK CLOCK DRIVERS FOR BASE BOARD FAN OUT



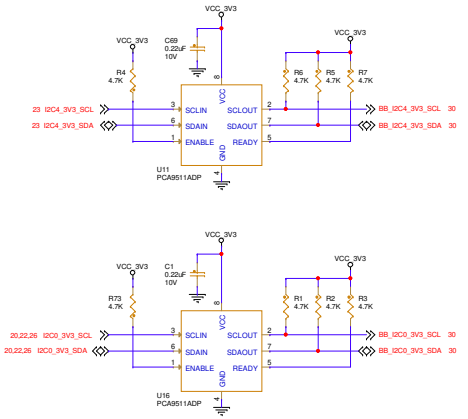
LEVEL TRANSLATOR



ETHERNET TX BUFFERS FOR BASEBOARD



I2C BUFFERS FOR BASE BOARD



ICAP Classification: CP: ___ IUO: ___ PUBI: X

Drawing Title: i.MX 8QM CPU CARD

Page Title: LEVEL TRANSLATORS & DRIVERS

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MISCELLANEOUS

GLOBAL FIDUCIALS



HEATSINK WITH FAN



TEST POINTS



MOUNTING HOLES



MOUNTING HOLE FOR M.2 CARD



ICAP Classification: CP: ___ IUO: ___ PUBI: X		
Drawing Title: i.MX 8QM CPU CARD		
Page Title: MISCELLANEOUS		
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